

# Refractive Gravity: An Ontological Picture and the Foundations of Mathematical Formalism

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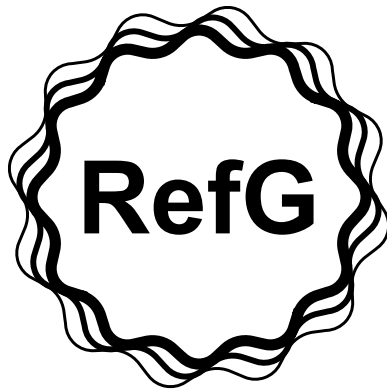
## Abstract

This monograph develops the ontological architecture of **Refractive Gravity** (RefG), a unified framework in which gravitation, matter, and the operational structure of measurement emerge from one universal pre-spatial foundation. Its Genesis scenario begins at a centerless initial boundary common to the entire foundation, assigned  $t = 0$  and  $\tau = 0$ , where the first distinguishable relations and physical event order appear. Oscillon seeds then localize within the manifested relational layer; overlap of their resonant fields increases network connectivity, while stable matter forms through later locking. Locally confined nonlinear wave configurations—*oscillons*—produce a static pressure deficit in the surrounding medium through Bernoulli-type dynamics, registered externally as active gravitational mass. In the static, spherically symmetric weak field, the common clock–ruler geometry gives the Parameterized Post-Newtonian (PPN) value  $\gamma = 1$ . The additive logarithmic response  $u = -\ln p$ , defined by a canonical gradient term and linear coupling to the active Gauss source, conditionally gives  $\beta = 1$  when no additional unsuppressed term enters at the same order. In the post-Genesis relational-coframe continuum, linearization on a regular Minkowski patch selects the Teleparallel Equivalent of General Relativity (TEGR) coefficient ratios by requiring the absence of an independent physical orientation mode within the leading local, reversible, constant-coefficient, parity-even quadratic-torsion class, with minimal phase coupling and no unsuppressed mixed operator at that order. The exact torsion–curvature boundary identity converts the selected action to Einstein–Hilbert form. Metric variation of its phase-current sector yields the Hilbert stress–energy tensor, while variation of  $\theta_C$  conserves the current. If the Einstein–Hilbert operator is the complete retained 1PN action, matter couples universally and minimally to its single metric, and no additional unsuppressed mode enters at that order, the complete standard PPN vector follows conditionally. Cosmologically, foundation expansion increases the inter-node scale  $a$ , lowers the mean pressure  $P_F$ , and thereby contracts the material scale-and-cadence factor  $p$ ; an internal observer reads this single causal history through  $A = a/p$ . Conservation of the shift-symmetry phase current, equivalently of its comoving charge, gives  $\eta_F a^3 = 1$  exactly, while homogeneous constitutive relations yield  $P_F/P_{F,0} = a^{-3}$ ,  $p = a^{-3/2}$ , and  $A = a^{5/2}$ . The thermodynamic pressure  $p_C$  describes the energetic state of the phase current, whereas  $P_F$  is the operational readout of foundation background pressure; the microscopic constitutive law relating them remains open. On the ideal homogeneous background, the same  $A$  determines redshift, signal time dilation, and geometric distances, matching the corresponding predictions of spatially flat  $\Lambda$ CDM at equal background parameters. A constant vacuum-phase contribution  $\Lambda_{\text{phase}}$  enters the single effective term  $\Lambda_{\text{eff}} = \Lambda_{\text{bare}} + \Lambda_{\text{phase}}$  once, with a positive total producing late operational acceleration. At microscopic scales, the volumetric-resonator, common-transposition, and  $C_3$ /Koide maps converge on one decisive physical bridge: a finite-energy oscillon family, its fluctuation spectrum, and an independently derived relation between oscillon energy and rest or pole mass.

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# Preface

This monograph presents an ontological account of Refractive Gravity (RefG), unifying diverse lines of theoretical inquiry into a single, coherent physical vision. RefG is a research program under stepwise, disciplined development. The central aim of this text is to provide the reader with a deep, intuitive picture of the fundamental architecture of the Universe; the underlying computational formalism and executable verification gates are preserved in the public scientific archive on Zenodo [Lotishvili, 2026a,b].

For the current status of the hypotheses, formal proofs, and empirical boundaries discussed throughout this work, the reader is referred to the concluding section, “Current Status of the Research Program.”

## Introduction

Modern fundamental physics rests upon a monumental, yet internally divided, architecture. On the one hand, Albert Einstein’s General Relativity describes gravitational interaction as the dynamic geometry of a smooth, four-dimensional pseudo-Riemannian spacetime; on the other, Quantum Field Theory represents matter and gauge forces as operator-valued excitations on a fixed, predominantly flat Minkowski background. Both pillars enjoy unprecedented levels of empirical confirmation, yet their internal ontological foundations remain fundamentally incompatible: gravity rejects a static background and dynamically generates the arena itself, whereas quantum mechanics, to preserve unitarity and its probabilistic structure, a priori requires a pre-determined, external temporal parameter.

At the interface of this profound disconnection, contemporary physics confronts a cluster of unresolved foundational crises:

1. **The Singularity Problem:** At the cores of black holes and at the cosmological origin, the continuous geometry of classical spacetime terminates in infinite curvature, signaling a breakdown of physical predictability.
2. **Astrophysical and Cosmological Anomalies:** Galactic rotation curves, the Radial Acceleration Relation (RAR), and the Baryonic Tully–Fisher Relation (BTFR) are conventionally accommodated by postulating dark matter halos, while the accelerated expansion of the Universe is attributed to dark energy (the cosmological constant  $\Lambda$ ), whose theoretical vacuum expectation value exceeds observational limits by 120 orders of magnitude.
3. **Structural and Hierarchical Puzzles:** The origin of quantum measurement randomness, the physical nature of the arrow of time, the pronounced mass hierarchy among charged leptons and quarks, the unexplained values of dimensionless coupling constants (such as the fine-structure constant  $\alpha \approx 1/137$ ), and the extreme weakness of gravity relative to the electroweak scale ( $M_{\text{Pl}}/m_e \sim 10^{22}$ ) remain arbitrary, disconnected inputs in the Standard Model.

The history of physics demonstrates that such persistent, isolated paradoxes typically indicate the absence of a deeper ontological substrate. Numerous grand unifications have sought to bridge this divide: Albert Einstein spent his final decades attempting to subsume electromagnetism and matter into an extended spacetime geometry; John Archibald Wheeler explored the nexus of geometry and information through the guiding principles of *Mass without Mass* and *It from Bit* [Wheeler, 1955, 1990]; Andrei Sakharov interpreted gravity as an elastic resistance (“induced gravity”) emerging from the quantum fluctuations of the vacuum [Sakharov, 1967]; Ted Jacobson derived Einstein’s field equations directly from the thermodynamic equation of state of local causal horizons [Jacobson, 1995]; Erik Verlinde formulated gravitation as an emergent holographic entropic force [Verlinde, 2011]; and Roger Penrose formalized the inescapable connection between global causal structure, gravitational collapse, and spacetime singularities [Penrose, 1965].

While these frameworks employ distinct mathematical vocabularies, they share a singular, foundational question: **Is it possible that the entire tapestry of physical phenomena—space, time,**

**gravity, and the particle spectrum—is the dynamic manifestation of a single primary ontological substrate?**

This monograph answers this question in the affirmative, presenting the unified conceptual and ontological framework of **Refractive Gravity** (RefG).

## Operational Foundations and Three Fundamental Principles

The standard language of physics typically begins with an established measuring apparatus: smooth coordinates, a continuous time axis, and operator fields are treated as axiomatic primitives. RefG grounds its inquiry in the strict operationalism of Percy Bridgman [Bridgman, 1927]: *Before speaking of space, time, or mass, what physical process generates the local, distinguishable differences upon which rulers and clocks function?*

In 1905, Einstein dismantled the concept of absolute time by operationally defining simultaneity via clocks and light signals [Einstein, 1905]. RefG pushes this operational critique to a more primordial level: **What physical mechanism constructs the stable clock, the material particle, and the propagating wave signal upon which all such measurements fundamentally depend?**

In response to this question, RefG formulates three fundamental postulates:

1. **Monistic Ontological Foundation:** Nature possesses strictly one universal, pre-spatial, and pre-temporal foundation. Its resonant self-distinction generates a manifested medium whose phase, pressure, shear, rotational, and topological modes are operationalized as elementary fields, forces, and particles.
2. **The Oscillon Nature of Matter:** A stable massive material unit is an *oscillon*—a locally confined, self-sustaining nonlinear wave configuration of the manifested medium. Massless carrier modes are the freely propagating radiation waves of the same medium. The localized oscillatory action of an oscillon generates a persistent, long-range static pressure-deficit profile in the surrounding substrate.
3. **Gravity as Effective Refraction:** Oscillon matter establishes an inhomogeneous pressure-deficit landscape throughout the medium. This deficit alters the effective geometry, causing the wavepaths of light and matter to bend refractively. Gravitational attraction is thus revealed as the physical mechanism of spacetime curvature. In the relational-coframe description, the pressure deficit is the volumetric scalar readout of the full local geometry. Symmetric shape and shear determine the tensor structure of gravity, while frame orientation is an inertial gauge freedom.

## Lorentz Invariance and Historical Precedents

The Michelson–Morley experiment of 1887 detected no “aether wind” from the Earth’s orbital motion, placing stringent constraints on 19th-century models of a rigid, matter-independent “luminiferous aether” [Michelson and Morley, 1887]. The RefG foundation-medium belongs to an entirely different ontological category: here, measuring rods, clocks, the observer, and propagating light are themselves collective wave excitations woven from this very medium.

When an observer and their apparatus are co-emergent wave modes of the medium, uniform translational motion corresponds to a uniform phase transformation of the wavepacket. Along the Lorentz-covariant branch, this motion leaves no dissipative wake or detectable “wind.” This realization revitalizes Lord Kelvin’s (William Thomson’s) visionary intuition regarding “vortex atoms” [Thomson, 1867]: **Matter is not a foreign body inserted into a passive background—matter is the organized, self-sustaining motion of the medium itself.**

## Seven Core Theoretical Contributions

This text establishes the ontological architecture of RefG, endowing the computational formalism archived on Zenodo with unambiguous physical meaning. While medium-based unifications have

notable precedents—such as Winterberg’s Planck aether model [Winterberg, 2002]—RefG places particular emphasis on the operational genesis of measurability and population-level resonant locking.

The core theoretical findings of this work are distilled into seven principal points:

1. **The Birth of Time and a Common Oscillon Cadence:** In the origin scenario of RefG, time begins at a centerless initial transition common to the entire foundation, which produces the first minimally distinguishable relations. Physical duration and event order are undefined on the pre-origin side of this boundary. Nonlinear modes localize as oscillon seeds only within the relational layer that has already formed, while stable material oscillons arise through later locking. In the common-cadence particle picture, a change in environmental pressure reorganizes the entire local resonant configuration: the coordinate frequencies of all particle modes change with the common cadence  $\Omega_t$ , while the oscillon’s internal structure determines the dimensionless labels  $s_i$  and their ratios.
2. **Two-Channel Mass Accounting:** The theory rigorously decouples two physical quantities: the oscillon’s internal topological state (its invariant internal energy store) and its externally read gravitational mass, which represents the volume-integrated pressure deficit induced in the surrounding medium.
3. **Biconformal  $p/p^2$  Scaling and the Complete 1PN Response:** In the first-order biconformal description, the external coordinate footprint of a local ruler, externally read mass, and clock cadence scale directly as  $p$ . Its coordinate period therefore scales as  $p^{-1}$ , while the coordinate speed of light defined by the same standard scales as  $p^2$ . One metric dictionary encodes this common response. In the static, spherically symmetric weak field, the common clock–ruler geometry gives  $\gamma = 1$ . The additive logarithmic response  $u = -\ln p$ , defined by a canonical gradient term and linear coupling to the active Gauss source, gives  $\beta = 1$  when no additional unsuppressed term enters at the same order. In the post-Genesis relational-coframe continuum, absence of an independent physical orientation mode determines the TEGR coefficients in the regular linear Minkowski limit of the leading local, reversible, constant-coefficient, parity-even quadratic-torsion class, with minimal phase coupling and no unsuppressed mixed operator. The exact torsion–curvature boundary identity converts this action to Einstein–Hilbert form. When this Einstein–Hilbert operator is the complete retained 1PN action, matter couples universally and minimally to its single metric, and no additional unsuppressed mode enters at that order, the complete standard PPN vector follows.
4. **The Koide  $C_3$  Phase Geometry:** A three-phase map in charged-lepton square-root mass space reproduces Koide’s  $K = 2/3$  identity exactly when  $A_K = \sqrt{2}$ . RefG reads this geometry as a candidate imprint of a ninth-order return cycle in which  $h = 2$  fixes the phase angle  $\theta = 2/9$ .
5. **The Nodal Origin of the Fine-Structure Constant ( $\alpha$ ):** The fine-structure constant  $\alpha \approx 1/137$  is interpreted as the dimensionless impedance ratio of the substrate’s electric-flux and transverse-photon response channels, remaining invariant under local pressure transformations.
6. **A Proposed Genesis Scenario, Foundation Expansion, and Its Internal Readout:** RefG describes the beginning of the Universe as a centerless boundary common to the entire foundation, assigned  $t = 0$  and  $\tau = 0$ . The first distinguishable relations arise there and physical time begins. Within the relational layer that follows, resonant fields of oscillon seeds overlap, resonant coupling increases network connectivity, and stable matter, radiation, and long-term relaxation emerge in later phases. In the post-Genesis realization, growth of the foundation’s inter-node scale lowers  $P_F$ , and the pressure decline reduces the material scale  $p$ . An organized neutral collective phase charge is distributed over the expanding three-dimensional measure. Conservation of its phase current gives  $\eta_F a^3 = 1$  exactly; homogeneous constitutive relations then give  $P_F/P_{F,0} = a^{-3}$ ,  $p = a^{-3/2}$ , and  $A = a/p = a^{5/2}$ . The phase-current thermodynamic pressure  $p_C$  describes the phase’s energetic state, whereas  $P_F$  is the operational readout of foundation background pressure.

The microscopic constitutive law relating  $p_C$  to  $P_F$  remains open. On the ideal homogeneous background, the same  $A$  determines redshift, signal time dilation, and geometric distances; at equal background parameters these observables match the predictions of spatially flat  $\Lambda$ CDM.

7. **Two Readouts of a Single Spectral Operator—A Central Hypothesis:** The short-wavelength eigenfunctions of a unified volumetric Helmholtz/Laplace-type spatial resonator are read as microscopic particle states, whereas its long-wavelength modes are read as candidate organizers of the cosmic web and galaxy-cluster structure.

## Historical Lineage and Priority Boundaries

Every component of RefG is anchored in the established heritage of theoretical physics:

- The  $p/p^2$  biconformal scaling shares conceptual kinship with the Polarizable Vacuum (PV) frameworks of Dicke and Puthoff [Dicke, 1957, Puthoff, 2002]; in RefG, it arises from the discrete ontology of a nodal network and two-channel mass accounting rather than phenomenological dielectric permittivity.
- The three-phase geometric reading of Koide’s formula extends the pioneering work of Brannen and Foot [Brannen, 2010, Foot, 1994], while the holonomic origin of  $2/9$  resonates with Shulga’s compact family cycles [Shulga, 2026]. RefG unifies this structure with  $\theta$ -blindness arguments and the  $C_3$  tetrahedral symmetry of a 3D volumetric resonator.
- Cosmological vacuum-pressure relaxation builds upon the  $q$ -theory of Klinkhamer and Volovik and its tendency toward saturation [Klinkhamer and Volovik, 2008], while the motivation for three spatial dimensions is related to studies of topological knotting and vortex-line stability [Whitrow, 1955, Berera et al., 2017].

The defining contribution of this monograph is the synthesis of these disparate threads into a single, logically connected research program: tracing the path from the operational birth of measurability and oscillon mass to galactic vortex dynamics and cosmological horizons.

The text invites the reader to follow this chain and see how a unified physical intuition unfolds into a broad panorama of contemporary fundamental physics.

## 1 The Ontological Foundation

### 1.1 A Single Foundation and Its Manifested Medium: Substance, Void, and the Dynamics of Existence

Modern fundamental physics describes the Universe as an assembly of mutually independent fields. In the Standard Model, each field is equipped with its own internal gauge symmetry, intrinsic degrees of freedom, and coupling parameters. Refractive Gravity (RefG) replaces this pluralistic ontology with a single, uncompromising monistic principle: **Nature possesses strictly one universal, pre-spatial ontological foundation.** Its resonant self-distinction generates a measurable, multi-channel medium. What we observe as distinct fundamental fields, gauge interactions, and elementary particles are the topological, phase, pressure, shear, and rotational modes of this single manifested medium. Throughout this monograph, the term “foundation-medium” denotes this self-distinguished, physically manifested state of the primary substrate.

This monistic vision possesses deep historical roots in theoretical physics. As early as 1870, William Kingdon Clifford formulated the prophetic hypothesis that matter consists of small, propagating ripples of spatial curvature [Clifford, 1876]. RefG endows this geometric intuition with a concrete physical mechanism: the self-distinguished medium supports not only geometric curvature responses, but also pressure, shear, hydrodynamic, and resonant channels.

A prominent theoretical precedent for the emergence of diverse particle and gauge sectors from a single substrate is the *string-net condensation* framework of Michael Levin and Xiao-Gang Wen [Levin and Wen, 2005]. In their model, collective many-body entanglement within a simple bosonic system naturally gives rise to both gauge bosons (photons, gluons) and fractional fermionic excitations (electrons, quarks). While the Levin–Wen model presupposes a pre-existing spatial lattice, RefG grounds this unifying capacity in the dynamic self-distinction of a single, pre-spatial foundation.

Conceptualizing the Universe as a unified continuum requires two asymptotic dynamical bounds:

1. **The Absolute Maximum of Substance:** A saturation limit characterized by maximal pressure, mass-energy potential, and operational nodal density.
2. **The Zero (Vacuum) Bound:** A limit in which interaction and physical measurability vanish entirely.

This “void” is not an abstract metaphysical nothingness, but the operational limit where physical distinguishability ceases. Objective physical reality unfolds as the bounded, nonlinear oscillatory dynamics between these two poles, wherein stable local units of the foundation—“nodes of existence”—are formed.

The working background  $\varphi = 0$  is the quiescent yet physically active state situated between these two extrema. At the limit of zero distinguishability, nodal interactions and measurable traces completely vanish;  $\varphi = 0$  designates the unperturbed baseline state of the manifested medium.

The ontological unity of the manifested medium is accompanied by a rich multi-channel capacity. The medium can compress (volumetric pressure), undergo topological twisting, establish vortical circulation, and support transverse wave excitations. A hydrodynamic analogy provides useful intuition: a single fluid concurrently supports isotropic hydrostatic pressure, vortex circulation, shear stress, and surface waves. Similarly, the RefG manifested medium integrates phase, pressure, shear, and topological responses, providing the dynamic substrate upon which electric charge, spin, and the particle spectrum are constructed.

The unperturbed state ( $\varphi = 0$ ) represents the baseline vacuum regime. The foundation is omnipresent; what varies locally is its degree of excitation, the operational density of distinguishable nodes, and the coupling strength between them. A drop in pressure, or rarefaction, corresponds to an increase in the unmeasured operational intervals separating nodes.

Here, “pressure” serves as the effective state variable of energy density in RefG. The baseline  $\varphi = 0$  allows deviations in both directions:  $\varphi > 0$  corresponds to a compressed, high-pressure branch, while  $\varphi < 0$  denotes a rarefied, deficit branch. The entire gravitational readout in RefG is constructed upon the gradient profile of this deficit branch ( $\varphi < 0$ ). Classical Bernoulli pressure serves as the direct physical analogue: for an external observer, a local shift in  $\varphi$  is simultaneously registered as gravitational time dilation, coordinate retardation of light, operational length contraction, and effective gravitational mass.

## 1.2 From Single Existence to Multiplicity: The Emergence of Measurability and Operational Space

Under the working postulate of RefG, physical change, event order, duration, and geometry are not yet assigned to the foundation’s initially undifferentiated state. In this picture there is no elapsed time before the birth of the Universe: birth establishes the first physical event and the origin of temporal reckoning.

The most coherent working origin scenario adopted here is a centerless initial transition common to the entire foundation. It is one initial condition with no distinguished spatial center. It does not describe a signal propagating through an already existing space, because physical distance, propagation speed, and an external vantage point are undefined on its pre-origin side.

At this boundary the foundation first produces minimally distinguishable relations. The physical ordering of events and both temporal measures begin there, with  $t = 0$  and  $\tau = 0$ . This first layer of



distinguishable connections is the minimal network of existence; material particles and stable oscillons have not yet formed.

Once these relations exist, some nonlinear modes localize as oscillon seeds. Resonant fields generated by those seeds propagate over the existing relational layer and overlap. In the working scenario, when a physical distinguishability threshold is crossed, the resulting resonant coupling joins operationally separated regions into one connected network. The first distinguishable relation therefore precedes this coupling. Local locking and network-wide connection may be partly contemporaneous; their actual ordering must follow from the dynamics of the resonant fields and the physical coupling threshold.

The most natural candidate form of the Universe is an interconnected network with neither a boundary nor a distinguished center. An exact finite periodic-graph construction shows only that finiteness, connectivity, and the absence of a unique center are mathematically compatible. It is an existence witness, not a derivation of the Universe’s dimension or topology.

Energy in the initial coherent state is distributed among oscillon seeds, their resonant fields, freely propagating waves, a thermal sector, and the collective deformation of the foundation. A continuity law accounts for this energy through a volume source, boundary flux, a central contribution, and the energy newly enclosed by the advancing activation boundary; exchange among the sectors cancels in the total balance. Phase energies and the equation of state determine the physical direction of that redistribution. The *melting* of ice is the intuitive image of this transformation.

In this scenario the first collective change initiates the foundation’s self-distinction. Operational space becomes measurable when a stable relation between nodes reaches the minimum distinguishability scale  $\ell_*$ . The interval between any two distinguishable nodes is the physical embryo of space; signal delay across their relation produces measurable duration, while the local energetic strain of differentiation is the ontological embryo of mass. Externally read gravitational mass is defined later by integrating the deficit profile that this strain induces in the surrounding foundation.

Physical meaning invariably requires comparison. At the ontological origin, three indivisible elements emerge:

- **Node:** A locally distinguishable, discrete state of the foundation;
- **Trace:** The energetic and phase alteration imparted by a node to the surrounding medium;
- **Relation:** The operational capacity to compare one local state with another.

From this foundational triad unfolds the entire operational network that macroscopically manifests as space, time, and matter.

A completely undistinguished foundation possesses a single, indivisible identity. Its resonant self-distinction generates the first persistent difference, marking the birth of quantity and counting. Distinguishable manifestations are registered as separate physical units without the foundation fragmenting into disconnected parts—multiplicity consists of the stable, mutually distinct states of one fundamental identity.

The primary unit of measurement is not an isolated object, but a preserved relation. In an undistinguished state, no second entity exists for comparison. Self-distinction yields two distinct states of the same foundation and establishes the primary link between them; here begin counting, spatial extension, and the relative ordering of events.

In information-theoretic terms, the first preserved distinction is analogous to a bit: a distinguishable trace is registered as “1”, and its absence within a given relational frame as “0”. Complex arrangements of these binary states generate rich macroscopic structures, just as the mutually distinct states of the foundation generate the diversity of the observable cosmos.

The concept of a single object existing at different locations at different times illustrates this logic. If the temporal labels along its worldline are removed, a single identity leaves traces across multiple locations. RefG elevates this intuition to the pre-spatial domain: persistent self-distinctions of the foundation construct the very locations that can subsequently be counted and compared. The atemporal

status of the foundation and the metric age of the observable Universe thus belong to distinct conceptual levels.

This perspective is crucial for understanding the nature of the observer. Observers and their measurement instruments are themselves complex, composite wavepackets built from this nodal network. This deepens Percy Bridgman’s operationalism, which asserts that physical concepts are defined solely by the operations required to measure them [Bridgman, 1927]. RefG provides this operational philosophy with an ontological substrate, investigating the physical conditions that make measurement operations possible in the first place.

The observable Universe is not matter placed inside an empty geometric container; it is the ongoing process of self-distinction within the foundation-medium. Operational space emerges along the traces of this distinguishability. Beyond these traces, the foundation may persist, but as a physical object it manifests only where it leaves an operational, measurable imprint.

The absolute separation between nodes cannot be measured directly, because no observer possesses a measuring rod external to the medium. What is measurable is strictly the operational trace of the inter-node interval. Operational space is born when the relation between at least two nodes becomes physically distinguishable. Below this threshold, variation remains an unmanifested internal state; above it, it registers as shifts in clock rates, operational sizes, mass readouts, and signal propagation.

RefG posits a dynamic lower bound of minimal operational distinguishability, denoted by  $\ell_*$ :

- At scales smaller than  $\ell_*$ , no independent measurable relation can form (information is indistinguishable);
- At the scale  $\ell_*$ , the first stable, physically distinguishable record emerges;
- At scales larger than  $\ell_*$ , inter-node intervals are operationalized as changes in pressure, clock rates, and signal propagation.

The physical rationale for this bound is that an infinitely fine record would lack dynamical stability. While the Planck length  $\ell_{\text{Pl}} = \sqrt{\hbar G/c^3}$  is a dimensional scale constructed from fundamental constants,  $\ell_*$  represents the dynamical threshold for stable record formation. Their potential coincidence depends on the microdynamics of the foundation.

Consequently, the ontological origin of the Universe is not a pre-existing manifold of geometric points. Prior to the emergence of inter-node intervals, there are neither distinct points nor spatial extension. Such a state cannot even be termed a “single point,” as a point is a concept belonging to an already realized geometry; it is a completely undifferentiated primary foundation-state. Multiplicity and geometry emerge only when an asymmetry generates a stable trace, rendering inter-node relations operationally readable.

Here lies the physical essence of gravity: when local coupling between nodes weakens (pressure falls), this state is simultaneously registered by an external observer as clock retardation, contraction of operational scale, and signal delay. This coordinated triad of physical responses is precisely what macroscopic observers perceive as a “gravitational field.”

### 1.3 Nodal Transmission: Waves, Time, and Self-Calibration

The foundation’s first collective change creates a minimal relational layer. Each node is a functionally distinguishable state that preserves the trace of a physical change and can relay it to adjacent nodes. Subsequent transmission, together with resonant coupling produced by overlapping resonant fields of oscillon seeds once the distinguishability threshold is crossed, joins this layer into an operational network. Spatial proximity and operational distance emerge from the stable organization of these physical couplings.

A wave in this network is the sequential transmission of state and phase. Light is a freely propagating transverse phase trace, whereas an oscillon is a locally confined, self-sustaining configuration of the same transmission mechanism. When matter moves, this confined wavepacket is transferred from node

to node, reconstructing its phase integrity at each successive location. Light and matter are thus the propagating and localized modes of a single medium.

In the working origin scenario, the first foundation-wide self-distinction creates the first ordering of events and begins time. Every subsequent transmission within the formed network extends that order. Recurrent phase relations establish a process cadence, while comparisons among stable oscillon cycles make clock measurement possible. The local rate of time is the network's collective rhythm: the rate at which the medium relays state and an oscillon completes its internal cycle.

Thermal insulation in double-pane windows provides a helpful analogy. Rarefying the intermediate gas reduces molecular collisions and slows the rate of heat conduction. RefG extends this intuition to the nodal network, linking the transmission rate directly to nodal density and coupling strength.

The observer's body, clocks, rulers, light, and matter are all modes of this same medium. In a deficit region, nodal density, transmission rates, oscillon cycles, and operational sizes undergo a coordinated shift. Local measuring devices reconfigure synchronously with their environment; consequently, an observer using local standards measures normal clock rates, standard dimensions, and the invariant local speed of light ( $c$ ). Comparing processes that have traversed different environments, however, reveals objective differences in accumulated time, operational size, and mass.

Along the selected Lorentz-covariant low-energy branch, this transmission is isotropic, non-dissipative, and strictly respects Lorentz symmetry: the local speed of light remains constant, and uniform motion leaves no frictional wake.

#### 1.4 Local Collective Cadence and Resonant Locking

The birth of time and the formation of a local collective cadence are distinct stages. In the Genesis picture of RefG, time begins with the first foundation-wide self-distinction. A local common cadence forms later, when the intrinsic frequencies of stable material oscillons settle into fixed, persistent ratios. In that collective state, the temporal rate is governed by the internal coordination and nonlinear mutual locking of the local network, in a manner analogous to Kuramoto-type synchronization.

At a deeper ontological level, the foundation possesses neither a preassigned tone nor a ready-made clock. After the connection network has formed, the manifested medium behaves like an acoustic instrument—a vibrating volumetric resonator whose connection operator and boundary conditions determine the admissible modes. The three-dimensionality of this resonator belongs to the physical map of RefG.

The physics of a bell offers a vivid illustration: its sound arises from the normal modes of one elastic body, arranged in a collective pattern of harmonic and anharmonic relations. Likewise, elementary-particle frequencies are bound by mutual balance and coordination within the common dynamics of the foundation. The universal foundation determines the arena of admissible modes, while particle modes balance one another within that arena to form a stable, resonantly locked configuration.

The pressure state changes the collective network cadence measured per unit foundation coordinate time. Denote that cadence by  $\Omega_t(x)$ ; the subscript  $t$  emphasizes that the frequency is counted per unit of foundation coordinate time. The common pressure response is

$$\frac{\Omega_t(x)}{\Omega_{t,0}} = p(x), \quad d\tau = p dt. \quad (1)$$

Here  $\tau$  is process time read by material clocks. A falling  $p$  means that fewer material cycles occur within the same coordinate-time interval  $dt$ . A local observer uses the co-adjusted  $\tau$  standard and therefore reads a normal local cadence; the difference appears only when environments or epochs are compared.

The oscillon's internal geometry selects its stable modes through dimensionless structural labels  $s_i$ . The common-cadence particle picture combines these modes as

$$\nu_i^{(t)}(x) = s_i \Omega_t(x), \quad \nu_i^{(\tau)} = \frac{\nu_i^{(t)}}{p}. \quad (2)$$

The  $s_i$  encode the oscillon’s internal modal structure, whereas  $\Omega_t$  is the environmental coordinate-time cadence. A common pressure change therefore transposes the entire spectrum, while process time removes that transposition from the local clock readout. In this description, the  $s_i$  turn the shared  $\Omega_t$  transposition into the distinctions among particle modes.

In a low-pressure region, the coordinate-time cadence  $\Omega_t$  decreases. Inter-node transfer is more frequent in a dense, high-pressure foundation and less frequent in a rarefied one:

- The retardation of inter-node signal transmission,
- The reduction in substrate pressure-energy density, and
- The reduction of the coordinate-time cadence  $\Omega_t$

represent three distinct operational readouts of a single ontological reality; within the formal mathematical chain, each enters exactly once.

Local clocks and atomic processes co-adjust with their environment, so an observer cannot detect the shift using local standards alone. Common transposition requires dimensionless ratios to remain invariant; the observed stability of the fine-structure constant is one of the sharpest tests of this principle. In this picture, the physical source of particle mass ratios lies in the internal  $s_i$  spectrum of oscillons.

The historical precursor to this relational approach is Ernst Mach’s principle [Mach, 1883]. RefG gives that principle a local, dynamical form: the common cadence of time is governed directly by the collective state of the local resonant network, while the influence of distant masses must be encoded in that local state. Relationality and local causality are therefore properties of the same local network.

In the RefG framework, the fine-structure constant  $\alpha \approx 1/137$  is interpreted as the dimensionless impedance ratio between the substrate’s electric-flux and transverse-photon response channels. On the common-transposition branch, it must remain invariant under local shifts in substrate pressure.

Now consider a wave signal propagating from one pressure regime of the foundation to another—for example, from a gravitational well to a distant observer. The photon follows the operational geometry  $A = a/p$ , while the receiver compares its phase cadence with a local atomic standard. On the ideal transparent branch this comparison gives

$$\frac{\omega_o}{\omega_e} = \frac{A_e}{A_o}, \quad 1 + z = \frac{A_o}{A_e} = \frac{a_o/p_o}{a_e/p_e}. \quad (3)$$

The common pressure response of the signal and the atomic standard is already contained in the  $A$  geometry; the resulting phase ratio is translated into the local  $\Omega_t$  cadence at each endpoint.

In summary, the rate of time in RefG is the ontological state of the local resonant network. When substrate pressure shifts, the network undergoes a collective rescaling: all compatible processes scale synchronously, leaving dimensionless ratios strictly invariant. An independent, abstract universal time possesses neither operational utility nor physical meaning: every physical measurement is executed strictly from within the network itself.

This principle establishes a rigorous empirical falsification criterion: **If any dimensionless ratio changes under a local pressure shift—for example, if a reliably established nonzero  $\Delta\alpha/\alpha$  is found—the postulate of a population-locked collective cadence is falsified.**

## 1.5 Oscillons, Pressure Deficits, and the Origin of Mass

What traditional physics designates as an “elementary particle”—whether an electron, proton, or quark—is recognized in RefG as an **oscillon**. An oscillon is a locally confined, self-sustaining, nonlinear topological wavepacket (a soliton-like state) supported by the foundation-medium. It is not an isolated, solid corpuscle distinct from the background; rather, it is a highly organized, coherent oscillatory state of the medium itself.

This concept continues the profound lineage of Lord Kelvin’s (William Thomson’s) “vortex atoms” [Thomson, 1867]. While 19th-century mechanical aether models were superseded, Kelvin’s core

insight—that particles are dynamic, self-confined topological modes of a continuous medium—remains a foundational concept that oscillon physics formalizes within modern nonlinear field theory and topology.

When an oscillon oscillates intensely, rotates, or confines energy within its localized geometry, the static pressure of the surrounding foundation-medium decreases. This behavior is illuminated by Daniel Bernoulli’s hydrodynamic principle [Bernoulli, 1738]: **An increase in internal dynamic intensity (velocity, oscillation amplitude) induces a corresponding drop in local static pressure.** On the weak, static gravitational branch of RefG, this hydrodynamic analogue corresponds to the schematic pressure-energy relation:

$$\Delta P \propto -\rho_{\text{dyn}}. \quad (4)$$

Through its continuous oscillatory action, an oscillon imprints a stable static pressure-deficit gradient upon the surrounding foundation. Within this framework, this pressure deficit is nothing other than the **direct external profile of gravitational mass**: integrating this deficit over all space yields the total gravitational mass of the system. The more intense the oscillon’s internal dynamics, the deeper the induced pressure deficit and the stronger the resulting refractive gradient. Simultaneously, the oscillon’s internal topological energy store is accounted for as a separate, decoupled physical channel.

The core of an oscillon is localized, but its oscillatory trace extends throughout the connected medium. The component phase-locked to the source establishes a static gravitational profile, requiring zero net outward energy flux. When the source undergoes rapid dynamic reconfiguration, the coherent oscillatory component released from this channel propagates outward as a gravitational wave, carrying energy and momentum.

The local depth of the deficit and the asymptotically measured total mass of a system are distinct physical quantities. The former characterizes central gradient and compactness; the latter integrates the active core, substrate dressing, binding energy, and radiated flux into a single asymptotic balance. Consequently, following a binary merger, the central deficit deepens while the total externally read mass decreases by the exact amount of radiated gravitational-wave energy.

This perspective shares structural parallels with John Archibald Wheeler’s concept of *Mass without Mass*, in which a “geon” represents a massive entity bound entirely by the self-energy of its gravitational and electromagnetic fields [Wheeler, 1955]. While classical Wheeler geons are unstable to radiative decay, RefG stabilizes oscillons through topological charge conservation, resonant phase locking, and the topological knot invariant  $Lk = Wr + Tw$ .

Consequently, the term “effective refractive mass” ( $m_{\text{eff}}$ ) designates the externally read gravitational mass of an individual oscillon within a specified background medium. Under adiabatic background variations, the oscillon’s topological structure, internal energy, and phase ratios preserve their identity, while its operational scale and externally read mass adapt to the new substrate state. Rapid reconfiguration represents an independent dynamical process that redistributes energy into radiation channels. The total asymptotic mass of an extended system requires a volume integration and is not simply the linear sum of uncoupled single-oscillon  $m_{\text{eff}}$  values. Through this decoupling, the origin of mass is revealed as the establishment of the medium’s refractive response.

At this foundational level, **gravitational attraction is the refractive bending of one oscillon’s wavepacket within the pressure deficit generated by another.** A relaxed string provides an intuitive picture of a gravitational well: locally diminished pressure-energy weakens the effective wave propagation stiffness of the substrate. The deficit profile defines the effective geometry, and any matter wavepacket bends along geodesics of minimal optical path length according to the eikonal equation.

This defines the central dynamical reciprocity proposed by RefG: **An oscillon creates a pressure deficit through its internal oscillation, while this deficit determines its effective mass, operational scale, and trajectory through the medium.** Matter dictates the pressure state of the medium, and the altered medium dictates the refractive geometry of matter.

## 2 Gravity as Refraction

### 2.1 Three Simultaneous Operational Shifts

As established in the preceding chapter, the inter-node separation within the foundation-medium is not directly accessible to external measurement; an observer can access only its operational, physical traces. Around an oscillon, this trace manifests as a localized depression in the pressure-energy state of the substrate: distinguishable nodes are rarefied, and the informational and wave coupling between them is attenuated. For an external, macroscopic observer, this single ontological state manifests as three simultaneous, coordinated operational shifts:

1. **Deceleration of the Coordinate Speed of Light:** In a pressure-deficit zone, nodes are rarefied and the rate of inter-node transmission drops. Consequently, the propagation of light and all other massless wave signals is retarded in coordinate description. The locally measured speed of light remains strictly invariant ( $c$ ) for a local observer; what changes is the effective propagation rate evaluated in the coordinate language of a distant observer.
2. **Contraction of Oscillon Operational Size:** While the oscillon’s intrinsic topological form (its nodal winding configuration) remains invariant, it is populated by a reduced volumetric density of interacting nodes within the rarefied substrate. In external observational description, this is registered directly as a contracted operational size, manifesting as a modified effective mass readout.
3. **Retardation of the Local Resonant Clock Cadence:** Real physical clocks are composed of oscillon systems (atoms, molecules) whose internal tick rate depends directly on local nodal coupling. In a rarefied environment, this coupling weakens, decelerating the clock rate in coordinate time relative to an asymptotic observer. Locally, the clock continues to measure its own cadence normally and invariantly.

A physical analogy is provided by double-pane vacuum insulation: evacuating the intermediate space reduces molecular collision frequency and suppresses the rate of thermal transport. In the RefG framework, this corresponds to the relation between nodal coupling strength and coordinate clock speed.

These three shifts are coordinated operational readouts of one underlying pressure deficit—not independent physical effects. Their quantitative relation establishes the foundation of biconformal geometry.

Nodal rarefaction does not generate a “secondary space.” The inter-node interval is not an independent physical substance; macroscopic “void” is operationalized entirely through the language of pressure variables. The sole objective manifestation of substrate rarefaction is the localized reduction of its pressure-energy state.

### 2.2 Quadratic Coupling and the Factor of Two

The decisive step is not to attach three independent laws to one another, but to read the same local standard in asymptotic coordinates. The external coordinate footprint of a local ruler and the coordinate cadence of its clock both scale as  $p$ . The corresponding coordinate period scales as  $p^{-1}$ ; hence the coordinate speed of light defined by one ruler interval per clock period scales as  $p/(p^{-1}) = p^2$ .

Denoting the dimensionless pressure factor by  $p$ , this biconformal scaling is formalized as:

$$\frac{L_{\text{coord}}}{L_0} = \frac{m_{\text{eff}}}{m_0} = \frac{\Omega_t}{\Omega_{t,0}} = p, \quad \frac{c_{\text{coord}}}{c_0} = p^2, \quad p = e^{\varphi/2}. \quad (5)$$

Here  $L_{\text{coord}}$  is the footprint of a local ruler in the distant asymptotic coordinate chart, not its locally measured proper length. For the local observer it remains one unchanged unit. The clock law gives  $d\tau = p dt$ , while ruler calibration gives  $d\ell = p^{-1} d\sigma$ . The temporal metric coefficient is therefore  $p^2$  and

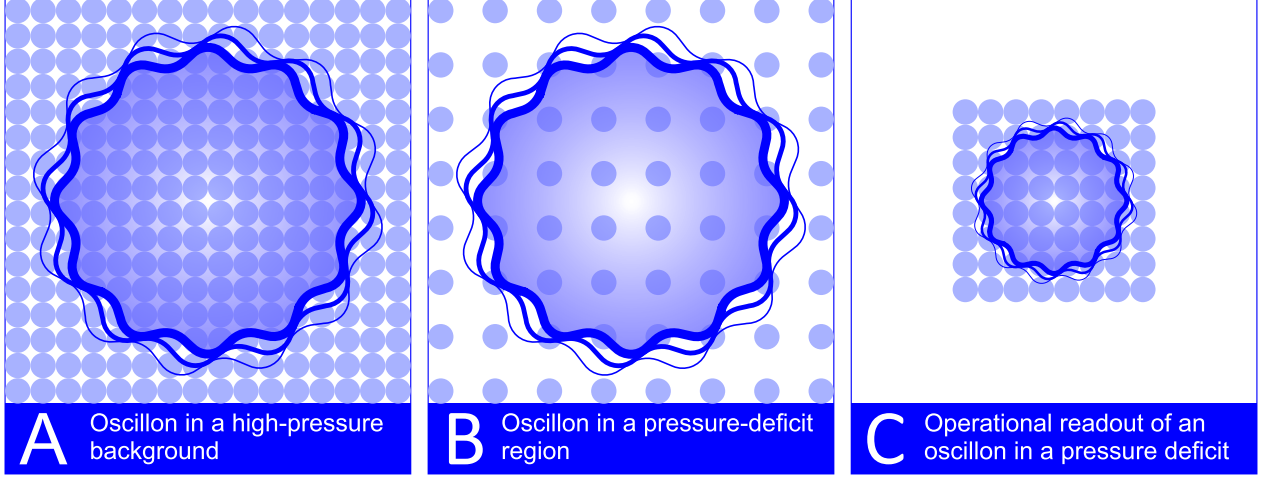


Figure 1: **Operational readout of an oscillon within a pressure-deficit zone.** (A) In an unperturbed, high-pressure foundation-medium, an oscillon resides in a dense nodal network; inter-node coupling is robust and local processes proceed at maximal cadence. (B) Within a pressure-deficit zone, substrate nodes are rarefied; operational inter-node distance increases, transmission rates decrease, and local temporal passage decelerates. (C) In external readout, an oscillon in a pressure deficit is registered with a slowed clock rate, contracted operational size, and shifted effective mass, while the internal separation does not enter as an independent direct observable.

the spatial coefficient is  $p^{-2}$ . A null signal has coordinate speed  $p^2 c_0$ , but reconstruction with the same local ruler and clock leaves the locally measured limiting speed exactly equal to  $c_0$ .

A similar power-law scaling appears historically in the Polarizable Vacuum (PV) models of Robert Dicke and Harold Puthoff [Dicke, 1957, Puthoff, 2002]. In RefG, these powers are assigned to distinct nodal-capacity and link-transmission channels rather than introduced through a phenomenological dielectric ansatz.

Formally, the external macroscopic readout of a single oscillon couples to the substrate pressure deficit through an exponential half-factor:

$$m_{\text{eff}} = m_0 e^{\varphi/2}. \quad (6)$$

On the deficit branch,  $\varphi < 0$ , confining the pressure factor to  $0 < p < 1$ . This relation describes the local external readout of an individual oscillon embedded within an inhomogeneous background: as  $\varphi$  deepens (pressure drops), its operational size and  $m_{\text{eff}}$  contract. Simultaneously, the total gravitational charge of the extended source generating the deficit is determined via a separate volumetric integration. This scaling follows directly from the theory's first-order biconformal branch.

Here, four physical quantities must be rigorously distinguished:

1. **Oscillon internal state and proper energy** ( $M_{\text{proper}}$ );
2. **Single-oscillon effective refractive mass in a given background** ( $m_{\text{eff}}$ );
3. **Active source profile of the induced pressure deficit** ( $M_{\text{active}}$ );
4. **Total asymptotic gravitational charge of the system** (ADM/Komar mass: Arnowitt et al. 1962, Komar 1959).

These four quantities organize into two nested accounting channels: the first distinguishes proper internal content from background-read mass; the second relates the externally read mass to the active core and substrate dressing. At the asymptotic boundary, these contributions combine into a single conserved charge, ensuring that every deficit trace is counted exactly once.

This self-consistent dynamic is captured by an intuitive physical analogy: **Imagine a crystal of colored ice dissolving in pure water.** The crystal is a locally frozen, ordered phase of the same water—just as an oscillon is a localized resonant locking of the universal foundation-medium. As the crystal melts, its dye disperses throughout the water, coloring the medium while the crystal itself shrinks in size. Energy is conserved: like the dissolved dye, localized binding energy transitions into an extended substrate pressure-energy state.

The more intensely the foundation is “colored” by this trace (i.e., the deeper the pressure deficit), the further local pressure drops, altering the externally read mass, operational size, clock cadence, and coordinate speed of light. This is a self-regulating relaxation process: the more trace an oscillon imparts to the medium, the lower its capacity for further emission. This negative feedback provides a natural mechanism for deficit saturation, with its precise boundary governed by full nonlinear dynamics.

This  $p/p^2$  scaling is formalized through the biconformal metric:

$$ds^2 = e^\varphi c^2 dt^2 - e^{-\varphi} d\sigma^2, \quad \varphi = -\frac{2U}{c^2}, \quad (7)$$

where  $U > 0$  is the gravitational potential of the selected exponential branch, coinciding at leading order with the Newtonian potential. In the weak-field limit ( $U/c^2 \ll 1$ ), the metric components expand in a Taylor series:

$$g_{00} = e^\varphi \approx 1 - \frac{2U}{c^2} + \frac{2U^2}{c^4} + \dots, \quad g_{ij} = -e^{-\varphi} \delta_{ij} \approx -\left(1 + \frac{2U}{c^2} + \dots\right) \delta_{ij}. \quad (8)$$

The common clock–ruler dictionary fixes the biconformal metric and its first-order spatial response. A conditional static functional determines the profile around a material source: its additive response is  $u = -\ln p$ , its gradient term is canonical, and its coupling to the active Gauss source is linear. Variation of this functional, together with the asymptotic boundary and Gauss charge, gives  $u = GM/(c^2 r)$ .

In this static, spherically symmetric weak-field description, the first post-Newtonian (1PN) metric expansion yields the Parameterized Post-Newtonian (PPN) coefficients:

$$\beta = 1, \quad \gamma = 1. \quad (9)$$

In this result,  $\gamma = 1$  is the first-order geometric consequence of the common clock–ruler response, whereas  $\beta = 1$  tests the second-order logarithmic response. Thus  $\beta = 1$  does not follow from the  $p/p^2$  kinematics alone; deriving it from the foundation Hamiltonian is the remaining microscopic task in this weak-field bridge.

**Relational Coframe, Einstein–Hilbert Geometry, and Full 1PN.** The covariant equations in this subsection use the  $(-+++)$  signature; the preceding biconformal presentation uses  $(+---)$ .

The static, spherically symmetric calculation tests the volumetric scalar response of the geometry. The complete local state is represented by a **relational coframe**  $e^A{}_\mu$ —a complete system of local frame one-forms. The coframe combines volume, shape, orientation, and the temporal direction and defines one operational metric,

$$g_{\mu\nu} = \eta_{AB} e^A{}_\mu e^B{}_\nu. \quad (10)$$

Its symmetric shape and shear carry the physical tensor response, while orientation is an inertial gauge freedom. This spacetime-coframe orientation is distinct from the physical internal three-axis structure of an oscillon associated with color in the particle map. Neighboring frames are compared by a flat, metric-compatible inertial connection, distinct from the electromagnetic  $U(1)$  connection. Failure of frame transport to close around an infinitesimal loop is **torsion**, the geometric measure of non-integrable deformation in the coframe continuum.

Linearizing the coframe response on a regular Minkowski patch shows that, within the leading local, reversible, constant-coefficient, parity-even quadratic-torsion class—with minimal phase coupling and no unsuppressed mixed operator at the same order—the absence of an independent long-range



physical orientation mode determines a unique coefficient ratio. The three independent quadratic torsion invariants are

$$I_1 = T^\rho{}_{\mu\nu} T^\mu{}_\rho{}^{\nu\nu}, \quad I_2 = T^\rho{}_{\mu\nu} T^{\nu\mu}{}_\rho, \quad I_3 = T_\mu T^\mu, \quad T_\mu = T^\nu{}_{\nu\mu}. \quad (11)$$

Their TEGR combination is

$$T_{\text{TEGR}} = \frac{1}{4}I_1 + \frac{1}{2}I_2 - I_3. \quad (12)$$

The selected low-energy action is therefore

$$S_F = -\frac{c_0^3}{16\pi G} \int d^4x e (T_{\text{TEGR}} + 2\Lambda_F) + S_C[g, J, \theta_C] + S_\partial, \quad (13)$$

where  $S_C$  is the covariant phase-current action and  $S_\partial$  is the matched boundary functional. The minus sign of the gravitational torsion term combines with the following torsion–curvature identity to recover the Einstein–Hilbert sign exactly. With the  $(-+++)$  signature, the exact connection identity

$$R_{\text{LC}} = -T_{\text{TEGR}} + \frac{2}{e} \partial_\mu (e T^\mu), \quad (14)$$

converts the coframe action to the Einstein–Hilbert action up to a boundary term [Beltrán Jiménez and Dialektopoulos, 2020, Maluf, 2013]. Here  $R_{\text{LC}}$  is the Ricci scalar of the Levi–Civita connection,  $e = \det(e^A{}_\mu)$  is the coframe determinant, and  $T^\mu$  is the torsion trace. Equivalence at the action level is exact for compact-support variations with suitable asymptotic falloff, or with the exactly matched boundary functional. Through this identity, Einstein geometry is the continuum representation of the relational-coframe law. The Fierz–Pauli/Deser spin-2 bootstrap supplies an independent crosscheck of the same final operator.

$S_C$  describes the phase current as a barotropic, isentropic, irrotational one-potential flow on its positive, stable, causal constitutive branch:  $\rho_C > 0$ ,  $\rho'_C > 0$ , and  $0 \leq n_C \rho''_C / \rho'_C \leq 1$ . Variation with respect to  $\theta_C$  gives the conserved current, whereas metric variation gives the Hilbert stress–energy tensor

$$T_{\mu\nu}^C = (\rho_C + p_C) u_\mu u_\nu + p_C g_{\mu\nu}. \quad (15)$$

[Brown, 1993]. Here  $p_C = n_C \rho'_C(n_C) - \rho_C(n_C)$  is the thermodynamic pressure of the phase sector, whereas  $P_F$  is the operational readout of foundation background pressure. Their relation belongs to an independent microscopic constitutive law. With the source restricted to the phase-current sector, the field equation is

$$G_{\mu\nu} + \Lambda_F g_{\mu\nu} = \frac{8\pi G}{c_0^4} T_{\mu\nu}^C. \quad (16)$$

For the general case, the right-hand side contains  $T_{\mu\nu}^{\text{tot}}$ ; every physical stress–energy sector in the action enters that total once through its own metric variation.

If, through retained 1PN order, the complete operational action consists of the Einstein–Hilbert action plus matter universally and minimally coupled to the same metric, the source is recorded by one conserved stress–energy tensor, RefG and GR use the same local source, PPN gauge, background, and boundary conditions, and no additional unsuppressed field or operator contributes at that order, then the complete standard PPN vector follows conditionally:

$$\gamma = \beta = 1, \quad \xi = \alpha_1 = \alpha_2 = \alpha_3 = \zeta_1 = \zeta_2 = \zeta_3 = \zeta_4 = 0. \quad (17)$$

This result is the conditional inheritance of the General-Relativistic 1PN response from action identity for isolated, weak-field, slowly moving, weakly self-gravitating sources. The static, spherically symmetric calculation provides a separate local check; within it,  $\beta = 1$  depends on the logarithmic law  $u = -\ln p$ .

A complete clock cycle is operationalized by the ratio of coordinate ruler footprint to coordinate velocity ( $\Delta t \sim L_{\text{coord}}/c_{\text{coord}}$ ). If a pressure drop halves that coordinate footprint ( $p = 1/2$ ) while

reducing coordinate light speed by a factor of four ( $p^2 = 1/4$ ), the period doubles—the clock runs precisely twice as slow.

In the shared static, spherically symmetric regime, the 1PN effects controlled by  $\beta = \gamma = 1$  recover the classical predictions of General Relativity: the gravitational deflection of light, Shapiro time delay, gravitational redshift, and planetary perihelion precession [Bertotti et al., 2003, Will, 2014]. In particular, the Cassini radio-link experiment measured  $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$ .

Historically, the additional **factor of two** arising from the spatial metric component  $g_{ij}$  was the decisive contribution that elevated Einstein’s 1911 half-deflection (derived solely from temporal dilation) to the full geometric result of 1915–1916 [Einstein, 1916, Shapiro, 1964]. In RefG, this factor of two emerges naturally from the differing ontological scaling of nodal volumetric density ( $p$ ) and link transmission delay ( $p^2$ ).

### 2.3 Refractive Attraction, Inertia, and the Absence of Friction

An oscillon, as an extended wave configuration, exists within the inhomogeneous gradient of substrate pressure and coordinate light speed. On the side where the medium is denser, wave phase propagates rapidly; on the rarefied (deficit) side, propagation is comparatively retarded. By virtue of the Huygens–Fresnel principle and the eikonal equation, the wavefront bends toward the region of lower pressure and reduced coordinate velocity—undergoing **refraction**. In the static weak-field limit, the pressure deficit generated by a massive body constructs an effective optical-geometric medium, and the refractive trajectory of a matter wavepacket within this medium is perceived macroscopically as gravitational attraction.

In RefG, the classic phenomenon of Newton’s falling apple is understood as follows: the oscillon wavepackets of the atoms comprising the apple bend refractively toward the region of lower pressure and slower coordinate propagation established by the Earth’s mass. Familiar gravitational attraction is not an occult action-at-a-distance force—it is the direct, local refractive response of matter waves to an inhomogeneous medium.

In the relational-coframe description, the pressure deficit and non-integrable deformation describe the state of the gravitational geometry. The scalar  $p$  profile gives the common volumetric response of clocks, rulers, and propagation, while the coframe carries the full tensor structure of the geometry. In the post-Genesis continuum, the TEGR action equals the Einstein–Hilbert action up to a boundary term; metric variation of the phase-current sector supplies the right-hand-side Hilbert source  $T_{\mu\nu}^C$ . Every physical stress–energy contribution present in the action enters the total Hilbert tensor once. A gravitational well is a region of depressed pressure and altered geometry, in which the total mass of a bound system is smaller than the sum of its separated constituents by the mass equivalent of the binding energy ( $\Delta M = B/c^2$ ).

The gravitational-wave event GW150914 provides a sharp observational benchmark for this energy balance: approximately three solar masses ( $3 M_\odot$ ) separating the initial binary mass from the final remnant were carried away as gravitational-wave energy [Abbott et al., 2016]. In RefG, two oscillatory sources merge into a single, deeper deficit profile. In the TEGR-equivalent low-energy geometry, their rapid nonlinear reconfiguration is read through the standard transverse-traceless tensor wave. Its node-scale shear spectrum, source–flux normalization, and complete waveform define the microscopic and source-model boundaries of the present account.

The conditional compact map for a regular core requires an effective violation of the Null Energy Condition (NEC). The stable phase current satisfies  $\rho_C + p_C = n_C \rho'_C > 0$  and belongs to the NEC-satisfying source class. A regular compact core therefore requires an appropriate equation of state and its unified source action.

RefG associates **inertia** with the dynamic deformation of an oscillon’s deficit profile during acceleration. When an oscillon is accelerated by an external force, its pressure profile develops a fore-aft asymmetry along the direction of motion; this gradient asymmetry generates a reactive resistance from the medium opposing the acceleration (the inertial force). Under deceleration, the profile reverses symmetrically.

Along a Lorentz-covariant, localized finite-energy branch satisfying the Laue condition ( $\int T^i_i d^3x = 0$ ), the inertial coefficient is determined by the total Noether energy of the system [Noether, 1918, von Laue, 1911]. Coupling this same energy as the universal source of gravity yields the equivalence  $m_i = m_g$  at leading order. The French satellite mission MICROSCOPE confirms the universality of free fall to a precision of  $10^{-15}$  [Touboul et al., 2022].

For an oscillon in uniform, rectilinear motion, the wave pattern is transferred symmetrically across substrate nodes, leaving no dissipative wake or net energy loss. RefG identifies **frictionless inertial motion** with this coherent phase-relay regime: motion is the wave-like relay of form. In the low-energy continuum description, the common Lorentzian cone relates the limiting rate of phase transmission to the metric light cone.

## 2.4 Local Unobservability and the Principle of Relativity

A local observer residing within a gravitational field reads clock rates and spatial dimensions as standard and unperturbed, because the observer’s clocks, measuring rods, and physical body are themselves constructed from the same foundation-medium. When substrate pressure shifts, the entire local physical system reconfigures synchronously. Relative to internal atomic transitions, slow variations in the gravitational field are strictly adiabatic: they preserve the vibrational mode and induce no spurious quantum transitions. Under this collective scaling, all dimensionless ratios—including the fine-structure constant  $\alpha$ —remain absolutely invariant.

This ontological picture is constrained by comprehensive experimental bounds:

- **Gravitational Redshift:** Confirmed with exceptional precision from the classic Pound–Rebka experiment [Pound and Rebka, 1960] to the hydrogen maser on Gravity Probe A [Vessot et al., 1980] and recent eccentric-orbit tests with Galileo satellites [Delva et al., 2018, Herrmann et al., 2018], verifying the local dependence of clock rates on the gravitational potential.
- **Invariance of Dimensionless Ratios:** High-resolution quasar absorption spectra and laboratory optical atomic clocks constrain cosmological drifts in the fine-structure constant to  $|\Delta\alpha/\alpha| \lesssim 10^{-6}$  [Kotuš et al., 2017]. RefG interprets this stability as consistent with local population-locked self-calibration.

The null result of the Michelson–Morley experiment reflects the medium’s **common-mode self-calibration**, whereas gravitational-wave detections by LIGO, Virgo, and KAGRA represent a **differential tensor response**:

- When the foundation-medium shifts uniformly and isotropically across a local domain, measuring rods, clocks, and optical propagation reconfigure synchronously in common mode, producing common-mode self-calibration in local measurements (modern optical cavities rule out spatial anisotropy in the speed of light to  $\Delta c/c \lesssim 10^{-17}$ ; Herrmann et al. 2009).
- A gravitational wave, by contrast, is a time-dependent, quadrupolar differential strain ( $h_+, h_\times$ ): it stretches space along one transverse axis while compressing it along the perpendicular axis. This differential deformation breaks common-mode cancellation, imprinting a measurable phase shift in interferometer arms.

FitzGerald–Lorentz length contraction [FitzGerald, 1889, Lorentz, 1892] is the historical precursor to this mechanism. RefG attributes this contraction to the fact that measuring instruments are themselves wave configurations of the medium.

In the TEGR-equivalent low-energy geometry, tensor waves propagate on the common metric light cone, so  $v_{\text{gw}} = c$  at this order. The simultaneous detection of gravitational waves and a gamma-ray burst from the binary neutron star merger GW170817 tightly constrains the fractional difference between gravitational and electromagnetic propagation speeds:

$$-3 \times 10^{-15} \leq \frac{v_{\text{gw}} - c}{c} \leq +7 \times 10^{-16} \quad (18)$$

[Abbott et al., 2017]. Concurrently, the orbital decay of relativistic binary pulsars (PSR B1913+16, PSR J0737-3039A/B) confirms general-relativistic quadrupole radiation damping to an accuracy of  $10^{-3}$ – $10^{-4}$  [Weisberg and Huang, 2016, Kramer et al., 2021], while bounds on scalar breathing modes from the GWTC-3 catalog tightly constrain scalar wave channels [LIGO Scientific Collaboration et al., 2022].

The most definitive operational invariant of this framework is the proper time difference accumulated by two clocks traversing distinct gravitational and kinematic worldlines:

$$\Delta\tau = \int \sqrt{g_{\mu\nu}dx^\mu dx^\nu}. \quad (19)$$

The elapsed time displayed on reunited clock faces is an objective, absolute physical record of distinct pressure histories, whereas instantaneously measured local rates and sizes remain relational readouts within the common substrate.

## 2.5 Historical Context and the Role of RefG

The refractive, optical-medium reading of gravity possesses an illustrious heritage. In 1911–1912, prior to completing the geometric formulation of General Relativity, Albert Einstein related the coordinate speed of light to the gravitational potential via:

$$c(\Phi) = c_0 \left( 1 + \frac{\Phi}{c^2} \right) \quad (20)$$

[Einstein, 1911, 1912]. Medium optics has even earlier roots: in 1851, Hippolyte Fizeau demonstrated that moving water partially drags light (governed by Fresnel’s dragging coefficient; Fizeau 1851), and in 1923, Walter Gordon proved that electromagnetic wave propagation in a moving dielectric medium obeys the null geodesics of an effective Riemannian metric [Gordon, 1923], establishing the first formal bridge between continuum optics and spacetime geometry.

In 1957, Robert Dicke formulated gravitation through a variable vacuum refractive index [Dicke, 1957]. The modern discipline of **analogue gravity** was inaugurated by William Unruh in 1981, demonstrating that sound waves in transsonic fluid flows propagate according to an effective acoustic metric exhibiting an event horizon [Unruh, 1981].

Grigory Volovik’s condensed-matter program [Volovik, 2003] and the extensive review by Barceló, Liberati, and Visser [Barceló et al., 2011] established how local Lorentz invariance and curved effective geometries emerge from the microscopic dynamics of quantum condensates (such as superfluid Helium-3 and Helium-4). A notable milestone was achieved in 2019, when Jeff Steinhauer’s group experimentally observed the acoustic analogue of thermal Hawking radiation in a Rubidium Bose–Einstein condensate [Muñoz de Nova et al., 2019].

Parallel to the medium-optics approach stands the **thermodynamic and entropic** school of gravity:

- Andrei Sakharov interpreted gravity as an elastic vacuum phenomenon induced by quantum fluctuations [Sakharov, 1967];
- Ted Jacobson demonstrated that Einstein’s field equations arise directly as an equation of state from the thermodynamic entropy balance ( $\delta Q = TdS$ ) across local Rindler horizons [Jacobson, 1995];
- Erik Verlinde formulated gravitational attraction as an entropic force driven by holographic information gradients [Verlinde, 2011].

These frameworks share with RefG the core conviction that spacetime geometry is emergent; in RefG, however, the effective geometry is generated not by abstract holographic screens, but by a concrete, pressure-refractive foundation-medium.

Refractive Gravity (RefG) introduces four fundamental ontological contributions to this historical continuum:

1. **Monistic Ontological Unity:** The foundation-medium is not merely a passive background for light; it generates matter itself—topological oscillons, measuring instruments, and observers. This moves beyond a passive-medium analogy and defines an independent, testable research framework.
2. **Local Population-Locked Cadence ( $\Omega_t$ ):** Nonlinear synchronization provides a candidate mechanism for local unobservability. The spectrum of one real oscillon must determine whether every stable mode is in fact transposed by one common factor.
3. **One Foundation with a Multicomponent Geometric Response:** Pressure and expansion govern the scalar readout, symmetric shape and shear carry the tensor response, and orientation is an inertial gauge freedom. Together these components form one operational geometry, and every physical stress–energy contribution present in the action couples to its metric once.
4. **From Relational Coframe to Einstein–Hilbert:** In the post-Genesis coframe’s leading local, reversible, constant-coefficient, parity-even quadratic-torsion class, with minimal phase coupling and no unsuppressed mixed operator at that order, the regular linear Minkowski limit determines the TEGR coefficient ratios by requiring the absence of an independent physical orientation mode. The exact torsion–curvature identity converts the action to Einstein–Hilbert form up to a boundary term [Beltrán Jiménez and Dialektopoulos, 2020, Maluf, 2013]. The Fierz–Pauli linear spin-2 construction and Deser’s universal self-coupling provide a separate route to the same final operator; Lovelock’s theorem independently establishes its geometric uniqueness [Deser, 1970, Lovelock, 1971].

In this historical landscape, RefG unifies optical refraction, topological solitons, and thermodynamic equilibrium into an integrated research program, with every link subject to formal mathematical and observational verification.

### 3 Three Scales and Cosmological Horizons

#### 3.1 The Local Scale: Compact Objects, Internal State, and External Readout

The universal pressure chain constructed throughout the preceding chapters manifests across three distinct astrophysical regimes:

1. **On the Local Scale:** As the localized pressure-deficit profile surrounding oscillons and compact relativistic bodies;
2. **On the Galactic Scale:** As the coherent, large-scale vortical flow and polarization of the foundation-medium;
3. **On the Cosmological Scale:** As the expansion-driven global relaxation of the foundation-medium and its two operational measures of time.

These three phenomena are distinct dynamical regimes of a single foundation. We begin at the local scale, where this mechanism reaches its most extreme expression: compact relativistic objects.

In extreme gravitational regimes, an object’s **internal state** ( $M_{\text{proper}}$ ) and its **externally read active gravitational mass** ( $M_{\text{external}}$ ) decouple fundamentally. The RefG compact map strictly distinguishes the invariant internal energy reserve of accumulated matter from its externally read gravitational charge. An external observer measures the system through its pressure-deficit profile, clock dilation, and operational scale contraction, whereas the interior geometry, horizon formation, and global causal structure are determined by the complete nonlinear field equations.

Two-channel accounting separates an object’s local internal content from its asymptotic gravitational readout. The internal channel tracks local matter states, conserved topological charges, and proper volume; the external channel registers the deficit profile, clock retardation, operational dimensions, and

total asymptotic mass. Gravitational compression sharply differentiates these two readouts without severing their common physical origin.

RefG attributes the mass defect of merging compact binaries to nonlinear reconfigurations in binding energy and the deficit profile. The quantitative connection to outgoing gravitational-wave flux is governed by the complete dynamical solution, benchmarked observationally by the binary black hole merger GW150914 [Abbott et al., 2016].

The biconformal metric governs the disparity between internal proper dimensions and external coordinate measures. On the compact branch of RefG, this divergence can become extreme: externally, such an object may appear ultra-compact, dark, and virtually “frozen,” while its internal proper volume remains vast. In both descriptions, the locally measured speed of light is invariant ( $c$ ); what differs is the temporal and spatial readout constructed by a distant observer. The “frozen star” appearance is an external observational feature, while the historical intuition of a “dark star” dates back to John Michell in 1784 [Michell, 1784].

RefG interprets a **singularity** as a boundary of external readout: under intense deficit conditions, the externally read value of the internal scale approaches zero, while the local proper scale halts at a finite saturation limit, preventing internal collapse to zero volume. A unified description requires that local energy density and spacetime curvature remain strictly finite.

One necessary premise in Roger Penrose’s singularity theorem [Penrose, 1965] is the relevant null energy/curvature condition (NEC:  $T_{\mu\nu}k^\mu k^\nu \geq 0$ ). The conditional regular-core map uses an effective violation of this condition. The stable phase current satisfies  $\rho_C + p_C > 0$ ; a non-singular interior therefore requires a physical source governed by a distinct equation of state.

Regular compact objects represent an active frontier in relativistic astrophysics. A prominent example is the Mazur–Mottola “gravastar”—a vacuum condensate object devoid of a singular core [Mazur and Mottola, 2004]. Such alternatives can be empirically distinguished from classical Kerr black holes via their Event Horizon Telescope (EHT) shadow morphology [Event Horizon Telescope Collaboration, 2019], quasinormal mode ringing, and late-time gravitational-wave “echoes” [Cardoso and Pani, 2019]. RefG shares structural kinship with this family, anchoring its regular interior in the pressure-energy balance of the universal foundation-medium.

### 3.2 The Galactic Scale: Vortical Flow and MOND

**Observational Reality.** In disk galaxies, rotation curves at large radii do not follow the Keplerian decline expected from visible baryonic matter alone: they exhibit rich morphological variety, frequently flattening or declining far more gently than Newtonian predictions. Concurrently, baryonic mass and asymptotic rotation speed obey a strict empirical law—the Baryonic Tully–Fisher Relation ( $M_b \propto v_f^4$ ; Tully and Fisher 1977, Lelli et al. 2016). This observational program began with Vera Rubin and Kent Ford’s measurement of the Andromeda galaxy [Rubin and Ford, 1970]. In the modern **Radial Acceleration Relation (RAR)**, 2,693 independent data points from 153 galaxies collapse onto a single universal curve [McGaugh et al., 2016].

The standard  $\Lambda$ CDM model accommodates these observations by postulating spherical dark matter halos. Modified Newtonian Dynamics (MOND), by contrast, introduces a universal acceleration threshold:

$$a_0 \approx 1.2 \times 10^{-10} \text{ m/s}^2, \quad (21)$$

below which gravitational acceleration transitions from Newtonian  $a_N$  to the deep-MOND regime  $\sqrt{a_N a_0}$  [Milgrom, 1983]. Relativistic completions (such as TeVeS and Skordis–Złóśnik models) incorporate additional dynamical vector and scalar fields [Bekenstein, 2004, Skordis and Złóśnik, 2021].

**The RefG Vortical Mechanism and Substrate Polarization.** At galactic scales, RefG attributes the additional deficit to a large-scale **vortical flow and polarization** of the foundation-medium. An ordered rotational regime deepens the central pressure deficit, while baryonic oscillons and the background flow form a single, shared refractive profile.

At the macroscopic level, this response is described by a Ginzburg–Landau-type local effective potential:

$$W(g_v, g_N) = \frac{1}{3}g_v^3 + \frac{1}{2}g_N g_v^2 - a_0 g_N g_v, \quad (22)$$

where  $g_v$  is the foundation-medium response field,  $g_N$  is the baryonic Newtonian acceleration, and  $a_0 \approx 1.2 \times 10^{-10} \text{ m/s}^2$  is the universal transition acceleration scale. The variational equilibrium condition,  $\partial W / \partial g_v = 0$ , yields:

$$g \equiv g_N + g_v, \quad g_N = \frac{g^2}{g + a_0}. \quad (23)$$

In the deep asymptotic regime ( $g \ll a_0$ ), this relation approaches  $g^2 \simeq a_0 g_N$ , yielding for circular orbits the Baryonic Tully–Fisher Relation ( $v^4 = GM_b a_0$ ) without individual galaxy tuning parameters.

**Observational Check on SPARC Data.** Distinct from the published 153-galaxy, 2,693-point RAR sample, the cuts  $\Delta V/V < 10\%$  and  $V_{\text{obs}} > 20 \text{ km/s}$  select 2,788 points from 163 SPARC galaxies. With fixed  $a_0$  and fixed mass-to-light ratios ( $\Upsilon_{\text{disk}} = 0.5$ ,  $\Upsilon_{\text{bulge}} = 0.7$ ), and without galaxy-by-galaxy parameter fitting, this comparison gives an RMS scatter of 0.141 dex and a mean residual of  $-0.012$  dex (Figure 2).

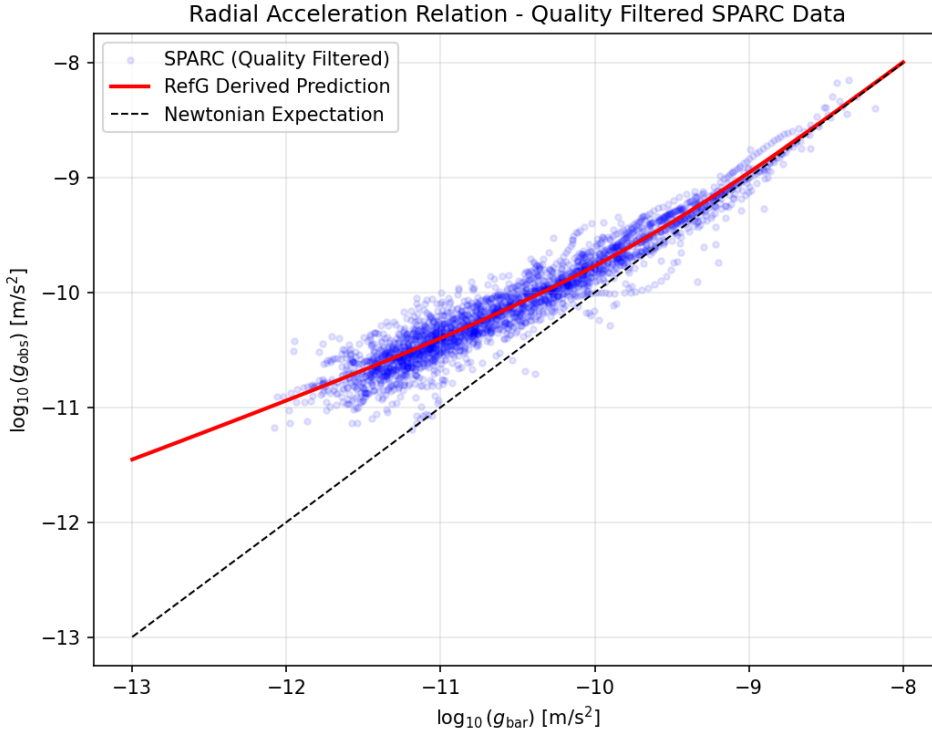


Figure 2: **Radial Acceleration Relation comparison with SPARC data.** The observed radial acceleration  $g_{\text{obs}} = V_{\text{obs}}^2/R$  versus the baryonic acceleration  $g_{\text{bar}}$  for 2,788 selected points across 163 disk galaxies. With fixed  $a_0$  and mass-to-light ratios, but no galaxy-specific fitting, the algebraic RefG curve obtained from  $W(g_v, g_N)$  gives an RMS scatter of 0.141 dex and a mean residual of  $-0.012$  dex.

**Disk Geometry and QUMOND Numerical Test.** For a finite-thickness Miyamoto–Nagai disk, transverse gradients of the algebraic field generate a geometric curl correction. On a  $300 \times 300$  grid with  $a = 3 \text{ kpc}$  and  $b = 300 \text{ pc}$ , Biot–Savart reconstruction gives, in this finite-grid configuration, a

maximum radial-acceleration correction of 9.23% relative to the one-dimensional algebraic mapping over  $a \leq r \leq 5a$ .

**Refractive Gravitational Lensing.** Along the flat rotation curve branch,  $\Phi_{\text{eff}} = v_f^2 \ln(r/r_0)$  and  $n(r) \approx 1 - 2\Phi_{\text{eff}}/c^2$ . Integrating the transverse refractive index gradient yields a deflection angle  $\hat{\alpha} = 2\pi v_f^2/c^2$ , matching the prediction of General Relativity for a singular isothermal sphere. Rotation and lensing thus emerge as two readouts of a single effective potential.

The closest modern analogue to this framework is the **Superfluid Dark Matter** model of Berezhiani and Khoury, where a MOND-like force arises from phonon excitations in a condensate [Berezhiani and Khoury, 2015]. However, while superfluid dark matter introduces an additional particle component, in RefG the coherent flow is a global regime of the unified foundation-medium itself.

**Morphology.** In RefG, galactic morphology reflects the flow regime of the foundation: ordered laminar rotation is associated with disk and spiral structure, whereas more chaotic or isotropic flow is associated with dispersion-supported elliptical galaxies.

**Galaxy Clusters.** In galaxy clusters, modifying the acceleration law alone does not fully resolve the dynamical mass deficit (historically highlighted by Fritz Zwicky in the Coma cluster; Zwicky 1933). RefG interprets the cluster mass budget through three distinct channels:

1. The global nodal deficit of the long-wavelength resonant network;
2. Individual galaxies and the long-range pressure-deficit fields of their constituent oscillons;
3. Cluster-scale vortical circulation and the hydrodynamic memory of past mergers.

All three enter a unified pressure-deficit budget, each counted exactly once.

Two decisive observational signatures distinguish this framework:

1. **Quantized Circulation:** If the substrate flow possesses a single-valued condensate phase, circulation is quantized in units of:

$$\Gamma = \oint \mathbf{v} \cdot d\mathbf{l} = n \frac{h}{m}, \quad (24)$$

where  $m$  is the effective mass of the flow mode. For ultralight modes ( $m \sim 10^{-22}$  eV), vortex filaments could be detected via weak lensing or stellar disk kinematics [Hu et al., 2000, Hui et al., 2021].

2. **The Frozen Memory of the Bullet Cluster:** In the Bullet Cluster (1E 0657-558), weak lensing centroids are displaced from the X-ray gas at  $8\sigma$  significance, tracking the collisionless galaxies [Clowe et al., 2006]. RefG interprets this separation as the **frozen memory of substrate vortical flow**: during collision, the gas was decelerated by hydrodynamic ram pressure, whereas the substrate deficit trace propagated forward with the galaxies. The memory timescale, spatial morphology, and lensing amplitude form a unified dynamical picture.

Thus, the unified observational program spans SPARC/RAR data [Lelli et al., 2016], cluster mass budgets, Bullet Cluster mergers, the CMB, BAO, and large-scale structure statistics.

### 3.3 The Cosmological Scale: Foundation Expansion and Its Internal Readout

The cosmological history of RefG begins with the centerless, foundation-wide transition described in Section 1.2. It is the common boundary of the first distinguishable relations. Distance and direction acquire physical meaning only after the first distinguishable connections have formed. In the centerless initial state, equivalent regions share one common boundary of birth; a history ordered outward from a



privileged site would break that initial symmetry. Local waves and oscillon centers arise only within the spatial layer that follows.

The sequence is: formation of the first relations, localization of oscillon seeds, overlap of their resonant fields, emergence of resonant coupling as the distinguishability threshold is crossed, growth of network connectivity, formation of stable oscillons, and long-term foundation expansion accompanied by pressure relaxation.

Four quantities carry distinct physical roles in this cosmological picture. The foundation background pressure  $P_F$  is an operational readout that controls the material scale-and-cadence response  $p$ . The quantity  $p_C = n_C \rho'_C(n_C) - \rho_C(n_C)$  is the thermodynamic pressure of the neutral phase current. The ordinary thermal pressure  $P_{\text{th}}$  is tied to physical temperature by an equation of state, while  $P_*$  denotes the mechanical response of a homogeneous stationary vacuum phase. Relations among these quantities are determined by their respective microscopic equations of state. A constant vacuum contribution enters the total  $\Lambda_{\text{eff}} = \Lambda_{\text{bare}} + \Lambda_{\text{phase}}$  once. The initial temperature is determined by the partition of energy among sectors, the equation of state for  $P_{\text{th}}$ , and energy exchange with radiation. Its observable content resides in dimensionless ratios such as

$$\Theta_\gamma = \frac{k_B T_\gamma}{E_{\text{atom}}}, \quad \mathcal{R}_i = \frac{\Gamma_i}{H_A}, \quad H_A \equiv \frac{1}{A} \frac{dA}{d\tau}. \quad (25)$$

A common  $p$  scaling cancels from these ratios, leaving the observable content of temperature to the real energetic relation between matter and radiation. The CMB spectrum, recombination, and BBN jointly test this thermodynamic map.

The quantity  $R_{\text{act}}(t)$  denotes a local post-origin activation or correlation scale within which nodal relations reach the minimum distinguishability threshold  $\ell_*$ . It is a local quantity; the homogeneous global geometry of the Universe is described by  $A$ . On a homogeneous background the same local rule applies around every node comoving with the network.

On the selected cosmological branch, formed matter follows the mean expansion of the nodal lattice. Let  $N_{12}$  denote the number of foundation units and links between two galaxies comoving with the mean flow. In the ideal expansion sector  $N_{12}$  is fixed while the foundation length of one link,  $\ell_F(t)$ , grows:

$$L_{12}(t) = N_{12} \ell_F(t). \quad (26)$$

This does not forbid local motion, rotation, collision, or gravitational displacement relative to the lattice; such processes may change  $N_{12}$ . It states only that the mean cosmological separation grows through the inter-node length scale.

The observer, the clock, and the ruler are made from the same foundation. Define

$$a(t) = \frac{\ell_F(t)}{\ell_{F,0}}, \quad (27)$$

and let  $p(t)$  be the pressure-controlled material scale-and-cadence factor. The selected branch contains one causal history,

$$a \uparrow \longrightarrow P_F \downarrow \longrightarrow p \downarrow, \quad (28)$$

which an internal observer reads through the single operational ratio

$$A(t) = \frac{a(t)}{p(t)}. \quad (29)$$

Foundation expansion, pressure relaxation, and contraction of the material standard are successive stages of one mechanism. The internal cosmological readout is the single ratio  $A = a/p$ , through which the foundation scale and the pressure response of the material standard enter one physical history.

The activation scale  $R_{\text{act}}(t)$ , the internal foundation scale  $a(t)$ , and the material standard  $p(t)$  are distinct quantities.  $R_{\text{act}}$  describes the local spread of stably distinguishable relations,  $a$  describes the change in inter-node length, and  $p$  describes the pressure response of material size and cadence.

**Neutral Collective Phase and the Relaxation Law.** A concrete post-Genesis realization of the relaxation law describes the organized foundation state by a neutral collective phase,

$$\Psi_C = \sqrt{n_C} e^{i\theta_C}. \quad (30)$$

Here  $n_C$  is the local density of organized collective phase charge, while  $\theta_C$  is its independent internal coordinate. Node number, energy, electric charge, process time, and the periodic phase of an oscillon belong to different physical roles and coordinates of the foundation.

The common shift symmetry of  $\theta_C$  generates a local current and conserves its total Noether phase charge  $Q_C$  within a fixed comoving region. Define the normalized spatial mean by  $\eta_F(t) \equiv \langle n_C \rangle / \langle n_C \rangle_0$ . On a homogeneous, isotropic background comoving with the network, the directional current vanishes. As the three-dimensional volume grows, the same  $Q_C$  is distributed over a larger measure, so  $\eta_F$  decreases:

$$\frac{Q_C}{Q_{C,0}} = \eta_F a^3 = 1. \quad (31)$$

A covariant phase-current action connects this conservation law to the post-Genesis coframe geometry. With the  $(-+++)$  signature, variation of  $\theta_C$  gives  $\partial_\mu J^\mu = 0$ , while metric variation gives the Hilbert stress-energy tensor,

$$p_C = n_C \rho'_C(n_C) - \rho_C(n_C), \quad T_{\mu\nu}^C = (\rho_C + p_C) u_\mu u_\nu + p_C g_{\mu\nu}. \quad (32)$$

This action describes a barotropic, isentropic, irrotational subfamily governed by one phase potential. The microscopic form of  $\rho_C(n_C)$ , its oscillon origin, and the constitutive relation between  $p_C$  and  $P_F$  remain to be derived.

Homogeneous cubic constitutive relations additionally set  $P_F/P_{F,0} = \eta_F$  and  $p^2 = \eta_F$ . Together with phase-current conservation, they give

$$\frac{P_F}{P_{F,0}} = a^{-3}, \quad p = \sqrt{\frac{P_F}{P_{F,0}}} = a^{-3/2}, \quad A = \frac{a}{p} = a^{5/2}. \quad (33)$$

Within this homogeneous map, expansion, pressure decline, and contraction of the material standard are successive readouts of one redistribution. The same conserved phase charge occupies an enlarging three-dimensional measure, so  $\eta_F$ ,  $P_F$ , and  $p$  decrease together and the cosmological response forms one causal trajectory.

**Local and global relaxation follow one common law in RefG.** Low foundation pressure in a strong gravitational environment reduces the externally read size of matter through the  $p$  scaling. For the mass of Earth, the Schwarzschild diameter is approximately 1.8 cm, comparable to a hazelnut. The same pressure-size-mass relation operates on the cosmological background: foundation expansion lowers the background pressure, and the pressure decline reorganizes the operational size of matter, its externally read mass, and its process cadence. Bodies, clocks, and rulers share that response and therefore read their own units as locally unchanged. This is one event in three successive forms—foundation expansion, pressure relaxation, and contraction of the material standard—whose complete internal readout is  $A = a/p = a^{5/2}$ .

**Refractive Geometry and Background Dynamics.** In the homogeneous description restricted to the phase-current source, the post-Genesis relational-coframe action takes Einstein-Hilbert form up to a boundary term, and variation of the induced metric gives

$$G_{\mu\nu} + \Lambda_F g_{\mu\nu} = \frac{8\pi G}{c_0^4} T_{\mu\nu}^C. \quad (34)$$

Here  $T_{\mu\nu}^C$  comes from metric variation of the phase-current sector, while variation of  $\theta_C$  gives current conservation. When additional matter and radiation sectors are included, the right-hand side contains

the total  $T_{\mu\nu}^{\text{tot}}$ . The operational  $A$  geometry takes Friedmann form on the spatially flat, homogeneous, and isotropic reduction. The phase-current action determines its conserved stress–energy source. A complete description of localized oscillons and free radiation requires their covariant actions and Hilbert tensors, each coupled once to the same metric. At this order  $\Lambda_F \equiv \Lambda_{\text{eff}}$  is one effective cosmological term, whose positive value produces late operational acceleration.

The relations  $A = a^{5/2}$  and  $d\tau = p dt$  display the same cosmological history in two languages:  $a$  follows the inter-node scale of the foundation, whereas  $\tau$  is time read by material processes. Redshift, the kinematic Hubble rate, signal time dilation, and geometric distances all arise from the single  $A$  geometry. On the ideal homogeneous background, these observables form one operational equivalence class with spatially flat  $\Lambda$ CDM at equal background parameters.

**The Time Bridge and a Finite Process History.** Process time counts accumulated material cycles, whereas foundation coordinate time follows the underlying order of events. Their bridge is  $d\tau = p dt$ : the factor  $p$  specifies how many material cycles fit within one coordinate-time interval. Both measures begin with the first distinguishable relation, at  $t = 0$  and  $\tau = 0$ .

In the early Universe, a large  $p$  places many internal cycles inside a short coordinate interval. When  $p$  is integrable, that abundance of cycles remains a finite process history. The Universe therefore has a beginning in both time measures, although coordinate and process clocks read its age through different cyclic standards. Each material structure begins its own process history at the epoch of its formation.

**Supernova Time Dilation.** In the cosmological picture of RefG, a larger early  $p$  means that material processes completed more cycles within the same interval of coordinate time than they do in the present background; the local  $\tau$  standard remains normal within its own epoch. On the ideal transparent photon branch, spectroscopic redshift follows directly from the metric and from comparison of the source and receiver’s local atomic standards:

$$\frac{A_0}{A_e} = \frac{a_0/p_0}{a_e/p_e} = 1 + z. \quad (35)$$

Neighboring null signals in the same metric independently show that an event lasting  $\Delta\tau_e$  in source process time is recorded by the present observer with duration

$$\Delta\tau_{\text{obs}} = \frac{A_0}{A_e} \Delta\tau_e = (1 + z) \Delta\tau_e. \quad (36)$$

Type Ia supernova light curves and spectral aging exhibit precisely this  $(1 + z)$  time dilation [Blondin et al., 2008]. In RefG, expansion of the foundation scale  $a$  lowers  $P_F$ , the pressure response changes  $p$ , and the entire history from source to receiver is read through the single operational geometry  $A = a/p$ .

Independent calculations in the  $A$  language of RefG and the  $z$  language of spatially flat  $\Lambda$ CDM give the same ideal redshift, time dilation, Hubble function, angular-diameter distance, and luminosity distance at equal background parameters. For these observables, the two descriptions belong to one operational equivalence class. Possible distinguishing signatures must be sought in source and clock microphysics, calibration, and inhomogeneous dynamics.

**Early Structural Maturity (JWST).** A supernova time profile compares signals from different epochs through photon propagation. The maturity of a galaxy records accumulated process history: mass assembly, successive stellar generations, chemical enrichment, and dynamical organization all require many internal cycles. A larger early  $p$  may provide more accumulated process time. This extra time does not by itself prove galaxy growth; an independent model of collapse, cooling, star formation, and feedback is required. The early massive galaxies observed by JWST are therefore a stringent test of this branch [Labbé et al., 2023].

**Two Observational Channels for One Mechanism.** Supernova time dilation and the maturity of early galaxies provide two distinct observational channels. The first measures photon propagation and signal-arrival intervals; the second probes the structural consequences of accumulated process time. Together they connect global relaxation of the foundation, the historical evolution of clocks, and the growth of matter.

**Ontological Consistency and the Arrow of Time.** Within a single-medium ontology, the familiar **vacuum-energy problem** becomes a problem of common energy accounting. If a manifested vacuum phase carries a constant operational energy density  $\epsilon_*$ , it produces the negative mechanical response  $P_* = -\epsilon_*$  and  $w_* = -1$  as the volume grows. Its geometric contribution is  $\Lambda_{\text{phase}}$ , and the operational geometry depends on one effective cosmological term,  $\Lambda_{\text{eff}} = \Lambda_{\text{bare}} + \Lambda_{\text{phase}}$ . Each physical contribution enters this total once through its own variational source. A fixed total energy stored in a comoving cell is diluted by expansion and has matter-like behavior,  $w = 0$ . The conserved quantity  $Q_C$  belongs to the phase-action sector, whereas the vacuum term belongs to the energy sector described by  $\epsilon_*$ . The  $q$ -theory program provides a historical motivation for this mechanism [Volovik, 2003, Klinkhamer and Volovik, 2008].

RefG relates the macroscopic **arrow of time** to the global direction of foundation relaxation. Like a colored crystal dissolving in water, energy dispersed through the medium does not spontaneously reconcentrate into a solid localized crystal. Such statistical irreversibility [Planck, 1899] may connect the thermodynamic, cosmological, and radiative arrows to one source. A complete derivation of entropy production and irreversible dynamics remains open.

### 3.4 Quantum Randomness and Two-Level Time

Two-level time provides a clarifying lens for quantum phenomena. The spacetime coordinates of a measurement event are recorded in oscillon (clock) time, while the selection of a specific outcome is associated with the deeper, pre-clock state of the foundation. This pre-clock level cannot be measured directly by clocks, because clocks are themselves emergent oscillon systems.

Bell's theorem and its experimental verifications rule out local hidden-variable theories [Bell, 1964, Hensen et al., 2015]. Meanwhile, the **no-signaling theorem** ensures that entanglement correlations cannot transmit superluminal messages.

RefG interprets this as **atemporal internal coordination**: within the internal language of the foundation, entanglement is not a physical signal propagating through space; in the metric language of clocks, it appears as an instantaneous statistical correlation between separated measurement outcomes.

Wheeler's **delayed-choice experiment** is understood in the same terms [Wheeler, 1978, Jacques et al., 2007]: a delayed measurement choice determines whether wave interference or which-way particle information is realized. RefG's two-level time describes this without retrocausal signaling, as an expression of the underlying phase integrity of the state.

Related concepts in modern physics include:

- The atemporal ground level of the **Wheeler–DeWitt equation** ( $\hat{H}\Psi = 0$ ; DeWitt 1967);
- The **Page–Wootters mechanism**, where time emerges from subsystem quantum correlations [Page and Wootters, 1983];
- **de Broglie–Bohm pilot-wave theory**, characterized by non-local phase coordination [Bohm, 1952].

This quantum framework is governed by two strict quantitative requirements:

1. **Born's Probability Rule** ( $P = |\psi|^2$ ): The model must derive the observed statistical distribution of measurement outcomes [Born, 1926];
2. **The No-Signaling Condition**: Internal phase coordination must never generate a transmissible signal that violates relativistic causality.

## 4 Oscillons and the Origin of Matter

### 4.1 Many Patterns in One Ocean

To build physical intuition, imagine a single continuous ocean and the diverse wave motions that arise within it. Most waves form, propagate across the surface, and quickly dissipate; under specific nonlinear conditions, however, certain configurations self-organize into stable, localized, self-sustaining structures:

- One may form a toroidal vortex ring, with fluid circulating continuously in a closed loop;
- Another may manifest as a stable spherical pulsation (soliton);
- A third may take shape as a helical, twisting wavepacket.

This hydrodynamic picture illustrates the foundational thesis of this chapter: **Electrons, protons, photons, and quarks are distinct resonant and topological patterns supported by a single universal foundation-medium.** The continuous medium itself oscillates: massive particles are locally confined modes of this oscillation, whereas photons are freely propagating transverse wave states.

A historical precursor is Tony Skyrme’s model, where nucleons (protons and neutrons) are described as topological solitons—“skyrmions”—of a nonlinear meson field [Skyrme, 1961]. RefG elevates this intuition to the fundamental ontological level: all such soliton-like structures are normal modes of a single universal foundation-medium.

The classical image of a rigid, pointlike particle gives way to a **continuous wave-medium ontology**: spatial extent is characterized by the vibrational amplitude and phase of the foundation, while a localized event registered in a detector represents the resonant coupling between this extended vibration and the internal oscillon modes of the detector’s atoms. RefG thus grounds the spatial amplitude and phase of the quantum wave function in an explicit ontological substrate.

In an atom, the “electron cloud” is an actual vibrational amplitude distribution rather than the rapid orbital transit of a point mass. The extended state maintains a coherent phase structure throughout space, and the localized point of measurement emerges through local resonant locking with the detector.

The single-electron double-slit experiment synthesizes two foundational realities: the wave amplitude depends simultaneously on the global geometry of both slits, yet each detection event on the screen manifests as a discrete, localized point. Akira Tonomura’s group directly recorded this progressive statistical accumulation using electron microscopy [Tonomura et al., 1989]. RefG interprets an atom’s stationary electronic state as a standing, confined wave mode of the medium rather than an accelerating classical point charge; consequently, electromagnetic radiation occurs only during transitions between discrete states.

The spatial spread of a wavepacket and the intrinsic structural form factor of a particle are distinct physical concepts:

- Elastic scattering probes charge and energy distribution form factors, while deep inelastic scattering reveals internal parton structure and distribution functions;
- The spatial extent of a propagating wavepacket does not enter these observables as an intrinsic physical size.

Empirically, the electron exhibits a pointlike electromagnetic form factor down to scales of order  $10^{-18}$  m [Navas et al., 2024]. A direct empirical test of the oscillon framework is whether the form factor computed from its internal topological charge profile reproduces this pointlike behavior across accessible momentum transfer scales.

### 4.2 Localized Confinement: Massive and Massless Modes

In the RefG particle framework, the defining ontological dividing line between massive and massless sectors is **localized confinement**.

**Massive Oscillons.** The energy of a massive particle is confined within a localized, self-sustaining configuration. When a wavepacket is trapped via vortex circulation or stable nonlinear pulsation within a finite spatial region, Bernoulli-type dynamics induce a persistent static pressure deficit in the surrounding medium.

Long-lived localized nonlinear pulsations are well-known in classical field theory as **pulsions and oscillons** [Bogolyubsky and Makhankov, 1976, Gleiser, 1994, Copeland et al., 1995]. RefG proposes that their stability arises from population-level resonant locking and conserved topological charges. In external readout, this locally confined energy manifests as inertia and effective gravitational mass.

While standard field-theoretic oscillons eventually decay through radiation leakage, RefG proposes nonlinear phase-locking with the surrounding particle population as the mechanism that suppresses radiative loss and selects long-lived particle states.

**Massless Carriers (The Photon).** A massless mode possesses neither a rest frame nor a localized static core—it propagates through the network at the limiting phase transmission velocity ( $c$ ). RefG interprets the photon as a freely propagating transverse phase mode.

Helicity and localized rotation are fundamentally distinct phenomena:

- The  $\pm 1$  helicity of a photon describes the spatial spirality (handedness) of transverse electric and magnetic phase vectors along the propagation axis, without forming a localized static vortex;
- A massive mode possesses a localized topological core and a static pressure-deficit well.

In Einstein–Hilbert geometry, every localized or radiative sector whose diffeomorphism-invariant action depends universally and minimally on the common metric enters the total gravitational source once through its Hilbert stress–energy tensor. The standard electromagnetic action includes photon energy and momentum in the gravitational source, while a propagating photon leaves no permanent localized well. The same rule includes the internal energy of composite particles.

Each particle sector requires a covariant action connecting its node-scale topological form to its Hilbert stress–energy contribution and the pressure-deficit geometry that contribution generates. Completing this map requires the photon, hadron, and other particle actions and their Hilbert tensors. Local topological form, spin, electric charge, color, and polarization retain their distinct quantum roles.

### 4.3 Self-Forming Resonators and the Two-Level Filter

Nonlinear wave equations admit an infinite diversity of mathematical solutions; nature, however, realizes a sparse, strictly defined spectrum of stable elementary particles. Why does nature select this specific discrete set?

RefG organizes this proposed selection mechanism through a **two-level dynamical filter**:

**1. Level One: Kinematic and Population Filter.** An acoustic handpan provides an intuitive analogy. While the resonant length of a guitar string is externally fixed by its wooden body, a three-dimensional oscillon must sculpt its own resonant cavity within the medium. On a handpan’s surface, a stable acoustic note unites the fundamental frequency, its octave, and the compound fifth into a harmonic  $1 : 2 : 3$  triad.

This triad is a structural rhythm: a self-forming resonator binds several mutually coherent modes into a single physical entity. In RefG, an oscillon’s pressure deficit alters the local geometry, which in turn filters viable modes: a stable mode must propagate through the substrate while locking into resonant coexistence with the surrounding particle population.

**2. Level Two: Energetic Stability Filter.** Any mode that passes the kinematic filter but possesses an energetically accessible decay channel rapidly disintegrates into lighter, stable ground states.

Hadronic resonances ( $\Delta, \rho, \omega$ ) illustrate this: with typical lifetimes of  $10^{-22}$ – $10^{-24}$  s, they appear in scattering spectra but cannot endure as asymptotic particles, shedding energy into stable channels (pions, protons).

Electron decay has never been observed; as the lightest charged lepton, it is protected by electric charge conservation. The extraordinary experimental stability of the proton ( $\tau_p > 10^{34}$  years) is linked in RefG to corresponding topological protection.

The mathematical language governing these filters is that of nonlinear **attractors**—stable phase-space configurations toward which dynamical trajectories asymptotically converge.

A stable particle represents a self-consistent equilibrium: the wave configuration carves a pressure deficit in the medium, and this deficit acts as a potential well that stabilizes and reinforces the configuration. In classical field theory, Sidney Coleman’s Q-balls are protected by conserved internal Noether charges [Coleman, 1985]. In RefG, stability is reinforced by nonlinear phase-locking across the entire population.

This collective synchronization requires no external coordinator: Christiaan Huygens’s classic observation of pendulum clocks synchronizing through a common wooden beam [Huygens, 1665], mode-locking in lasers, and Yoshiki Kuramoto’s nonlinear synchronization model [Kuramoto, 1975] demonstrate how collective order emerges endogenously. In particle physics, this vision resonates with Geoffrey Chew’s S-matrix “Bootstrap” program, where particle spectra form a self-consistent whole [Chew and Frautschi, 1961].

Under adiabatic gravitational changes, resonant locks are preserved: the collective cadence  $\Omega_t$ , operational scale, and externally read mass adjust synchronously, while dimensionless ratios and phase harmonies remain invariant. Rapid, high-energy shocks can break this locking, redistributing energy into other compatible channels.

**From One Oscillon to the Particle Spectrum.** One localized oscillon family is the minimal complete physical unit of the particle picture. Its stable finite-energy form creates the spatial profile of a self-forming resonator; its small fluctuations reveal the internal  $s_i$  spectrum and connect it to the common environmental cadence  $\Omega_t$ . The collective phase  $\theta_C$  belongs to the organized collective state of the foundation, whereas the periodic oscillon phase belongs to its local cycle. Keeping these coordinates distinct places cosmological relaxation and particle rhythm within one coherent picture. The oscillon’s energetic content and the response it leaves in the surrounding medium form the physical bridge to rest and pole mass.

#### 4.4 The Global Resonator: Unity of Micro- and Macro-Spectra

The two-level filter specifies the proposed selection mechanism; we now ask where it is realized.

Ernst Chladni’s classic acoustic experiment provides the guiding intuition [Chladni, 1787]: sand scattered across a vibrating metal plate collects along the nodal lines of standing waves, where the oscillation amplitude is minimal. In the RefG map, this image corresponds to global modes and their nodes. Every resonator possesses its own spectrum: its geometry and boundary conditions determine which modes are amplified and which are suppressed.

After the operational network has formed, RefG treats it as a candidate **unified global volumetric resonator**. Only at this stage does the concept of an eigenspectrum acquire a defined meaning: the connection operator, network geometry, and boundary conditions select the admissible modes. The three-dimensionality of this resonator and the identification of its spectrum with physical particles and the cosmic web remain central hypotheses. During subsequent expansion and thermal evolution, its modes may lock into stable physical forms:

- **Short-Wavelength Modes** at high frequencies correspond to localized elementary particles, or oscillons;
- **Long-Wavelength Modes** at low frequencies correspond to the large-scale distribution of matter, the **Cosmic Web**.

According to the central hypothesis, the distribution of galaxies and galaxy clusters may resemble sand on a Chladni plate: filaments and vast cosmic voids may reflect the nodal pattern of standing modes in the global resonator. This identification requires a dedicated spectral test.

Standard cosmology describes this large-scale morphology through Yakov Zel’dovich’s anisotropic gravitational collapse [Zel’dovich, 1970]. RefG relates the same morphology to the ontological language of a single spectral operator.

The central hypothesis of this map is that **the microscopic elementary particle and the macroscopic cosmic web are two different readouts of one universal spectral operator**: one is a locally confined topological form, and the other is a globally extended cosmic pattern.

A joint empirical test of this operator must encompass:

1. The localized particle mass spectrum;
2. The cosmological history of structure formation;
3. The mass budgets of galaxy clusters;
4. The Baryon Acoustic Oscillation standard ruler at approximately  $100 h^{-1}$  Mpc [Eisenstein et al., 2005].

Within the candidate picture of a finite, boundaryless resonator, conceptually related to Christof Wetterich’s variable-gravity framework [Wetterich, 2013], cosmological expansion increases the foundation inter-node scale, lowers its background pressure, and thereby rescales its internal material standards. The resonator supplies its own geometry, clocks, and rulers.

In current CMB observations, the low-multipole anomalies at  $l = 2, 3$  remain moderate. Searches for cosmic topology establish topology-dependent lower bounds and do not select a unique global form [Planck Collaboration, 2016].

#### 4.5 Discreteness and the $C_3$ Generation Symmetry

In the Standard Model, the masses of the charged leptons—electron ( $e$ ), muon ( $\mu$ ), and tau ( $\tau$ )—are empirical Yukawa-coupling inputs. In the candidate map of RefG, they form the **ninth-order internal three-phase ( $C_3$ ) triad** of a single topological defect: three distinct resonant modes of one underlying physical structure.

The charged-lepton three-phase parametrization is written most cleanly in square-root-mass space. Defining  $x_k \equiv \sqrt{m_k}$  gives

$$x_k = \mu \left[ 1 + A_K \cos \left( \theta + \frac{2\pi k}{3} \right) \right], \quad k = 0, 1, 2, \quad (37)$$

where:

- $\mu$  is the common scale in  $\sqrt{m}$  space;
- $A_K$  is the dimensionless three-phase modulation amplitude;
- $\theta$  is the phase angle of this algebraic mass map;
- $k = 0, 1, 2$  indexes the three generations ( $e, \mu, \tau$ ).

When  $A_K = \sqrt{2}$ , the  $C_3$  algebra identically recovers Yoshio Koide’s empirical mass relation [Koide, 1982, 1983]:

$$K = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3}. \quad (38)$$

This three-phase parametrization possesses a rich literature:



- Carl Brannen formulated the relation in terms of circulant matrix eigenspaces [Brannen, 2006];
- Robert Foot demonstrated that  $K = 2/3$  corresponds to a  $45^\circ$  geometric orientation in  $\sqrt{m}$ -space [Foot, 1994];
- Brannen’s Pancharatnam–Berry phase construction yielded  $\pi/12$  while leaving  $\theta = 2/9$  as a dynamical parameter [Brannen, 2010];
- Konstantin Shulga obtained  $\theta_\ell = -2/9$  within a compact family cycle [Shulga, 2026].

RefG connects this structure to a **ninth-order holonomic return cycle**. On the selected  $h = 2$  branch, the first non-trivial topological return gives  $\theta = h/9 = 2/9$ , while  $A_K = \sqrt{2}$  expresses equal energy sharing between the internal singlet and doublet vibrational channels.

A simple  $C_3$ -invariant potential exhibits  **$\theta$ -blindness**—an inability to select  $\theta$  dynamically—indicating that the origin of the mass angle must be sought beyond a simple potential minimum, in locked topological holonomy.

Evaluating  $(A_K, \theta) = (\sqrt{2}, 2/9)$  at tree level, anchored to the physical electron pole mass, yields deviations of  $451.7\sigma$  for the muon mass and  $0.61\sigma$  for the tau mass. Reconciling this tree-level geometry with measured pole masses requires an action-derived radiative correction bridge [Sumino, 2009].

This structure has a clear acoustic analogue: the charged-lepton  $C_3$ /Koide triad can be read as a slightly anharmonic chord of a three-dimensional volumetric resonator. In this interpretation, an energy–mass map connects the  $x_k$  variables to the oscillon eigenspectrum and measured pole masses. Here  $\theta$  is the family mass angle, whereas  $\theta_C$  is the collective phase of the foundation.

## 5 Topology and Quantum Numbers

### 5.1 Phase, Orientation, and Topological Quantization

Thus far, the foundation-medium has been described primarily in terms of vibrational amplitude and local pressure. However, the complete local state of the medium is characterized by multiple coupled field variables: the phase-clock scalar, amplitude profile, and spatial orientation frame each possess distinct degrees of freedom. In this monograph,  $\Phi$  represents the compact designation for the complete state of the foundation-medium rather than a simple single-component scalar: its full configuration simultaneously carries amplitude, phase, internal geometric form, and spatial orientation.

If  $\Phi$  were merely a single-valued scalar field, its velocity field would be defined strictly by the gradient  $\mathbf{u} = \nabla\Phi$ , forcing the curl (vorticity) to vanish identically ( $\nabla \times \mathbf{u} = 0$ ). The emergence of a non-trivial topological sector requires:

- **A multi-valued phase,**
- **A singular defect nodal line, and**
- **A localized internal spatial orientation frame.**

At the central core of a defect line, the wave amplitude vanishes ( $|\psi| = 0$ ), while around it the phase winds by an integer multiple of  $2\pi$  ( $\oint \nabla\theta \cdot d\mathbf{l} = 2\pi n$ ). A canonical laboratory analogue is a quantized vortex filament in superfluid Helium.

The cosmological analogue is a **cosmic string**—a macroscopic topological defect line formed during a symmetry-breaking phase transition in the early Universe via the Kibble mechanism [Kibble, 1976]. In RefG, this represents a “frozen seam” of the foundation: vacuum topology determines whether such lines form, while astrophysical observations (CMB lensing and pulsar timing arrays) tightly constrain their cosmological tension.

Lord Kelvin’s seminal papers on vortex atoms and vortex motion [Thomson, 1867, 1869] inspired Peter Guthrie Tait to systematically catalog knots, laying the foundations of modern mathematical

knot theory [Tait, 1877]. RefG continues this lineage using the modern vocabulary of pressure-energy balance and population-locked resonance.

History provides a crucial lesson: 19th-century vortex atoms in ideal inviscid fluids failed to yield a stable discrete spectrum or reproduce atomic spectra. In RefG, the topological knot provides the kinematic scaffolding of form, while physical viability and dynamical stability are secured by pressure-energy feedback and nonlinear phase-locking across the oscillon population.

Phase quantization occurs when the field is single-valued and continuous along a closed loop, returning to its initial state modulo  $2\pi$ , with the loop belonging to a non-trivial first homotopy group ( $\pi_1 \neq 0$ ).

For a closed, orientable, framed ribbon, the **Călugăreanu–White–Fuller (CWF) theorem** applies [Călugăreanu, 1961, White, 1969, Fuller, 1971, Ricca and Nipoti, 2011]:

$$Lk = Wr + Tw, \quad (39)$$

where:

- $Lk$  (Linking Number) is the topological integer invariant measuring total linking;
- $Wr$  (Writhe) measures the 3D spatial coiling of the ribbon’s central axis;
- $Tw$  (Twist) measures the internal phase winding of the ribbon frame around its axis.

Under continuous deformation,  $Wr$  and  $Tw$  can vary continuously (e.g., spatial coiling converting into internal twist), yet their algebraic sum  $Lk$  remains strictly invariant.

In fluid dynamics, kinetic helicity ( $H = \int \mathbf{u} \cdot (\nabla \times \mathbf{u}) d^3x$ ) is an invariant of ideal barotropic flow, which Keith Moffatt connected to the topological linking number  $Lk$  [Moffatt, 1969]. In superfluid Helium, circulation quantization was predicted by Lars Onsager and Richard Feynman [Onsager, 1949, Feynman, 1955] and experimentally verified by Joe Vinen [Vinen, 1961]. Integer phase winding is thus an established laboratory reality.

The core principle “**Topological Confinement = Conserved Charge**” implies in RefG that electric charge is the external readout of a closed phase-orientation winding: topological invariants define the discrete charge class, while field dynamics govern its electromagnetic coupling.

In the RefG synthesis, **spin-1/2 and elementary electric charge** are read as two candidate geometric traces of a common double-covered topological source: the half-twist determines spinorial transformation under rotation, while chiral orientation and integer winding label the charge class and sign. Their quantitative completion requires separate derivations of the spin transformation law and electric-flux normalization.

Direct experimental analogues exist: Kelvin waves on quantized vortices in superfluid Helium have been directly excited and their dispersion measured [Minowa et al., 2025]; in classical water vortices, helical axial modes and their dispersion relations have been mapped experimentally [Barckicke et al., 2026]. In RefG, an oscillon’s radial breathing mode corresponds to its Compton mass pulsation, while its helicoidal mode governs charge and radiative channels.

## 5.2 Spin and Fermionic Statistics

A phase field twisted by a half-turn ( $\pi$ ) along an oscillon’s internal axis generates a **Möbius-strip-type double-covered  $Z_2$  topology**:

- A single spatial rotation of  $2\pi$  reverses frame orientation, flipping the sign of the wave amplitude ( $\psi \rightarrow -\psi$ );
- A double rotation of  $4\pi$  fully restores the initial quantum state ( $\psi \rightarrow +\psi$ ).

This  $Z_2$  geometry embodies the fundamental transformation property of the **spinor representation** ( $SU(2)$ ). RefG associates electron spin-1/2 with this internal double-covering. Changing this topological class requires tearing the configuration: vanishing amplitude at the core, defect reconnection, or altering boundary conditions.

Completing the connection to Fermi–Dirac statistics requires that exchanging two identical oscillons produces a phase shift of  $\pi$ , yielding an antisymmetric multi-particle state in Fock space:

$$\Psi(\mathbf{r}_1, \mathbf{r}_2) = -\Psi(\mathbf{r}_2, \mathbf{r}_1). \quad (40)$$

Under the required exchange antisymmetry, the amplitude for two identical fermions occupying the same state vanishes identically ( $\Psi = -\Psi \implies \Psi = 0$ ). Deriving this exchange phase from the Möbius topology is the remaining step needed to recover the **Pauli Exclusion Principle**. For bosonic modes, the Möbius half-twist is absent, the state remains symmetric under exchange ( $\Psi \rightarrow +\Psi$ ), and macroscopic Bose condensation is permitted.

In relativistic quantum field theory, the Spin-Statistics Theorem unites Lorentz invariance, positive energy, and microcausality [Pauli, 1940]. RefG’s topological framework offers a candidate geometric mechanism for this connection.

### 5.3 Charge, the Photon, and the Electromagnetic Channel

RefG defines **electric charge** as the external spatial trace of an oscillon’s internal chiral phase twist:

- Right-handed chiral twist corresponds to positive charge ( $+e$ );
- Left-handed chiral twist corresponds to negative charge ( $-e$ ).

Externally, this trace extends radially in all directions. In quantitative electrodynamics, its surface integral must satisfy Gauss’s Law:

$$\oint \mathbf{E} \cdot d\mathbf{S} = \frac{Q}{\epsilon_0}. \quad (41)$$

A **positron** is the exact topological mirror image of an electron (possessing opposite chiral twist). In electron-positron annihilation ( $e^+ + e^- \rightarrow 2\gamma$ ), opposite topological windings untwist and cancel, releasing locally confined energy into freely propagating transverse radiation (photons). In a neutral oscillon, opposing chiral twists cancel internally, yielding zero net flux (characterizing the neutrino sector).

A uniform rotation of the internal phase angle corresponds to global  $U(1)$  gauge symmetry. Describing spacetime-dependent rotations necessitates a local  $U(1)$  phase connection and covariant derivative. RefG identifies the **photon** as a **transverse, helicoidal wave** of this connection: it propagates phase-twist disturbances across space, acting as the massless gauge boson of electromagnetism.

The topological nature of electromagnetism is manifest in classical electrodynamics: Antonio Rañada’s vacuum solutions exhibit linked, knotted electric and magnetic field lines classified by the Hopf invariant [Rañada, 1989].

The sign of physical forces (attraction vs. repulsion) is governed by gauge boson spin, source type, and coupling structure:

- **Vector Gauge Channel (Spin-1):** Like charges repel, opposite charges attract;
- **Tensor Gravitational Channel (Spin-2):** In the one-metric weak-field geometry, positive nonrelativistic sources couple with the same sign and attract at leading order; the response to relativistic pressure, stress, and vacuum sources is determined by the full  $T_{\mu\nu}$ .

In three spatial dimensions, a conserved massless flux dilutes as  $1/r^2$ , yielding a  $1/r$  potential.

Hydrodynamic Bjerknes forces [Bjerknes, 1906] illustrate the limits of simple fluid analogies: in an acoustic medium, bodies pulsating in-phase attract, while out-of-phase bodies repel—the exact opposite

of electrostatics. This demonstrates that force signs are not reducible to crude mechanical collisions; they are governed by the rigorous mathematics of vector gauge fields, source currents, and topological linking.

The fine-structure constant  $\alpha \approx 1/137$  sets the dimensionless coupling strength of this electromagnetic channel and is interpreted as the impedance ratio of the substrate's electric-flux and transverse-photon response channels.

## 5.4 Color Confinement and Three Generations

The three-dimensionality of space ( $D = 3$ ) plays a crucial structural role in the RefG particle map: **The three-axis spatial orientation frame of an oscillon provides the geometric substrate for the color charge triad (red, green, blue:  $r, g, b$ ).** Color represents the internal spatial degree of freedom of this frame:

- **Quark:** An oscillon mode vibrating along a single internal axis is topologically open, possessing unconfined endpoints (an open topological segment);
- **Baryon (Proton, Neutron):** The symmetric combination of all three axes ( $r + g + b$ ) forms a fully closed, stable 3D topological knot (a color-singlet white state);
- **Meson:** Pairing an axis with its opposite anti-axis (color + anticolor) likewise forms a closed, color-neutral loop.

RefG identifies the candidate mechanism for **color confinement in Quantum Chromodynamics (QCD)** with the geometric incompleteness of an isolated quark: a closed, gauge-invariant state cannot be formed from a single open axis.

The mode count of an unconstrained three-axis deformable oscillon matches the 8 generators of  $SU(3)$ , establishing the structural map of the color sector. Formal identity requires verifying the  $SU(3)$  structure constants, local gauge invariance, and interaction vertices.

The **Three-Generation Puzzle** is a proposed extension of the  $C_3$  structure explored in Chapter IV:

- Charged leptons ( $e, \mu, \tau$ ) represent the ninth-order three-phase vibrational modes of a single topological defect;
- Quark generations ( $u/d, c/s, t/b$ ) replicate this generational architecture across the three-axis colored sector.

In this candidate map, the three generations are read as successive holonomic return modes of a single 3D volumetric resonator.

The universal foundation of holonomy is **Berry's geometric phase** [Berry, 1984]: under adiabatic cyclic transport across parameter space, a quantum state acquires an invariant geometric holonomy phase alongside its dynamical phase. RefG links this to the ninth-order return cycle of a framed defect, yielding  $\theta = 2/9$  rad on the  $h = 2$  branch.

**Why is Space Three-Dimensional?** In RefG,  $D = 3$  is intimately tied to the **stability of topological knots**:

- Closed one-dimensional loops form stable, non-trivial knots and links exclusively in  $D = 3$  spatial dimensions;
- In higher dimensions ( $D \geq 4$ ), any knotted loop can be untied and unlinked smoothly along extra spatial dimensions without self-intersection.

This topological argument builds upon the insights of Paul Ehrenfest, Gerald Whitrow, and modern investigators [Ehrenfest, 1918, Whitrow, 1955, Berera et al., 2017]. RefG formulates the co-emergence of material discreteness and 3D space from a single topological locking mechanism as a central working hypothesis.

## 5.5 Geometric Map of the Standard Model

RefG organizes the particle spectrum of the Standard Model into the following geometric architecture:

**1. The Electron and Charged Leptons.** The electron is a complete toroidal, framed Möbius  $Z_2$  topological defect with all three spatial axes locally closed. RefG connects this double-covered geometry to the Dirac factor  $g = 2$ .

The electron's measured magnetic moment slightly exceeds 2. Julian Schwinger calculated the leading quantum electrodynamic correction ( $\alpha/2\pi$ ), while modern Penning trap experiments measure  $|g_e|/2 = 1.00115965218059(13)$  [Schwinger, 1948, Fan et al., 2023]. This precision measurement provides a benchmark for testing the quantum completion of the topological map.

**2. Quarks and Color Charge.** Quarks are open axial topological segments. Their fractional electric charges ( $+2/3, -1/3$ ) are associated with one-third subdivisions of the 3D closed loop; their exact values remain to be fixed by canonical charge normalization. Color confinement dictates that only closed, color-singlet configurations (baryons and mesons) exist as asymptotic states.

**3. Gauge Bosons.**

- **Photon:** Massless propagating transverse helicoidal wave ( $U(1)$  gauge symmetry, 2 helicities);
- **Gluons:** 8 massless transverse modes mediating color transitions between internal axes ( $SU(3)$  octet);
- **$W^\pm$  and  $Z^0$  Bosons:** Locally confined, massive carrier oscillons mediating electroweak transitions ( $SU(2)_L \times U(1)_Y$ ).

RefG interprets **beta decay** ( $n \rightarrow p + e^- + \bar{\nu}_e$ ) as a topological reconnection: altering the internal axis orientation ( $d \rightarrow u$ ) produces a transient charged weak carrier ( $W^-$ ) that decays into leptonic channels, bringing chirality, energy-momentum balance, and the decay spectrum into a single geometric transition.

**4. Neutrinos and the Higgs Boson.**

- **Neutrinos:** Electrically neutral, incompletely confined chiral phase modes. Their incomplete topological confinement is associated with their ultra-small masses ( $m_\nu < 0.1$  eV) and weak coupling, while Pontecorvo flavor oscillations represent phase interference among mass eigenspaces;
- **Higgs Boson:** A purely spherical, radial breathing pulsation of the foundation-medium devoid of angular momentum, naturally corresponding to a spin-0 scalar state ( $m_H \approx 125.25$  GeV).

**5. The Graviton and the Weinberg–Witten Theorem.** The TEGR-equivalent low-energy geometry yields the massless spin-2 sector of General Relativity and its two transverse-traceless polarizations,  $h_+$  and  $h_\times$ . The Fierz–Pauli/Deser spin-2 bootstrap independently checks the same result. Their emergence from the foundation's node-scale shear spectrum remains to be derived.

The Weinberg–Witten theorem [Weinberg and Witten, 1980] applies to a Lorentz-covariant quantum field theory on a fixed Minkowski background with a local, gauge-invariant microscopic  $T_{\mu\nu}$ . The pregeometric foundation of RefG begins from a different structure, while the Lorentzian metric and Hilbert source emerge in the low-energy operational continuum. The graviton is read as a collective tensor-shear excitation of the foundation-medium. A future microtheory satisfying every premise of the Weinberg–Witten theorem would provide a direct falsifier of the composite-graviton scenario.

**6. Nonlinear Soliton Lagrangian.** The formal mathematical foundation of this topological map relies on nonlinear field Lagrangians admitting stable knotted soliton solutions (in the spirit of the Faddeev–Niemi model; Faddeev and Niemi 1997).

## 5.6 Fundamental Constants: $\alpha$ and the Gravitational Hierarchy

The fine-structure constant:

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} \approx \frac{1}{137.035999} \quad (42)$$

remains one of the deepest mysteries in physics, described by Richard Feynman as one of the greatest unexplained numbers in science [Feynman, 1985].

In RefG,  $\alpha$  is interpreted as the **dimensionless impedance ratio of the substrate’s electric-flux and transverse-photon response channels**.

This definition is governed by three precise distinctions:

1. Simple ratios of Planck and electron masses or frequencies do not yield 137;
2. The identity  $r_e/\bar{\lambda}_c = \alpha$  is an algebraic consequence of dimensional definitions;
3. Integer phase-locking steps represent physical realities (as demonstrated by the Josephson effect and Shapiro steps; Josephson 1962, Shapiro 1963), though seconds are themselves calibrated by atomic oscillons.

Thus,  $\alpha$  is not an absolute frequency or energy; it is the **dimensionless impedance ratio of the substrate’s electric-flux and transverse-photon response channels**. Under local pressure shifts, both responses scale synchronously, leaving  $\alpha$  invariant.

**The Hierarchy Problem: Why is Gravity So Weak?** The electrostatic repulsion between two protons exceeds their gravitational attraction by a factor of  $10^{36}$ , while for electrons this ratio reaches  $10^{42}$ :

$$\frac{F_{\text{em}}}{F_g} = \frac{e^2}{4\pi\epsilon_0 G m_e^2} = \alpha \left( \frac{M_{\text{Pl}}}{m_e} \right)^2 \sim 4.17 \times 10^{42}. \quad (43)$$

This immense disparity motivated Paul Dirac’s Large Numbers Hypothesis [Dirac, 1937].

RefG interprets this hierarchy through the **differing ontological character** of the two interactions:

1. **Electromagnetism** is a primary, direct transverse-phase gauge coupling between topological charges;
  2. **Gravity:** An oscillon’s internal oscillatory energy induces a **secondary, Bernoulli-type static pressure deficit** in the surrounding foundation. Its long-range refractive trace is read as the gravitational field ( $-\Delta P \propto \rho_{\text{dyn}}$ ).
- The Planck mass ( $M_{\text{Pl}} = \sqrt{\hbar c/G} \approx 2.18 \times 10^{-8} \text{ kg} \approx 1.22 \times 10^{19} \text{ GeV}$ ) is the limiting calibration scale for the elastic response of the foundation-medium;
  - Electrons and protons are localized oscillon modes whose energies lie many orders of magnitude below that scale ( $m_e/M_{\text{Pl}} \sim 10^{-22}$ ).

Because the gravitational pressure deficit is proportional to oscillon energy ( $m_e$ ), and gravitational attraction couples two such deficits ( $Gm_e^2$ ), gravitational interaction between elementary particles is quadratically suppressed relative to the Planck scale ( $m_e^2/M_{\text{Pl}}^2 \sim 10^{-44}$ ).

In RefG, the extreme weakness of gravity is read as a difference between these two channels: gravity is a pressure-mediated refractive effect generated by microscopic oscillons. The ratio  $\alpha(M_{\text{Pl}}/m)^2$  reproduces the correct scale; its full origin requires a common dynamics for  $G$ ,  $M_{\text{Pl}}$ , and the microscopic energy of an oscillon.

## 6 Epilogue: Boundaries of Knowledge

### 6.1 Why One Foundation and One Manifested Medium?

At the conclusion of this work, we return to the central physical and philosophical question: **Why a single foundation-medium?**

One primary motivation is ontological parsimony (Occam’s razor): a single substrate radically minimizes the number of independent, disconnected axiomatic inputs. However, mathematical simplicity alone is insufficient to establish physical truth.

A far deeper and more decisive argument is the **common origin of physical law**. When electrons, quarks, and photons are understood as distinct resonant and topological modes of a single universal foundation-medium, RefG seeks to derive the conservation of energy, momentum, angular momentum, and electric charge from the dynamic equilibrium of this common substrate rather than postulating them as external, unconnected axioms. Physical interactions represent couplings between differing modes of the same medium, while conservation laws express the geometric continuity of its total energy balance.

This perspective addresses Eugene Wigner’s famous reflection on the “unreasonable effectiveness of mathematics” in the natural sciences [Wigner, 1960]. The postulate of a single ontological substrate provides this profound harmony with a physical explanation: **The universal mathematical order of physical law reflects the internal dynamical coherence of a single primary substance.**

Postulating multiple mutually independent fundamental fields (as in the Standard Model) merely defers the question: *What overarching symmetry principle constrains these disparate fields to obey the same universal conservation laws and relativistic spacetime symmetries?* Within the RefG ontology, this universality is not an external decree—it is the intrinsic dynamics of the shared substrate.

This argument gives the Standard Model’s **fine-tuning problem** a common structural target:

- The stringent observational bound on the electrical neutrality of bulk matter ( $|q_p + q_e|/e < 10^{-21}$ );
- The observational compatibility of the photon rest mass with zero ( $m_\gamma < 10^{-18}$  eV);
- The precise quantization of fractional quark charges [Navas et al., 2024].

These interrelated constraints call for a common structural explanation, which RefG associates with the topological geometry of the foundation-medium.

A single universal foundation gives rise to immense phenomenological richness: it concurrently supports volumetric pressure, shear stress, phase, topology, vortical circulation, memory, and resonance. This reflects Philip Anderson’s celebrated insight—“**More is Different**”: every emergent level of structural complexity exhibits its own autonomous physical laws [Anderson, 1972]. RefG understands transitions across scales as the self-organization of a multi-channel medium.

RefG reads the measurable cosmological Universe as a **self-measuring global resonant network**. The foundation itself is pre-spatial and is not assigned a spatial size. A mathematical witness exists in the form of a finite, connected periodic network with no unique center, but this witness does not determine the actual dimension or topology of the Universe. A finite, boundaryless global resonator remains a physical candidate. The Universe measures itself through its own resonant modes.

### 6.2 Three Operational Boundaries

Measurement of the Universe is always performed by components of the Universe itself: our clocks, measuring rods, detectors, and bodies are oscillon structures of the universal foundation-medium. This concept of a **self-measuring Universe** resonates with John Wheeler’s *It from Bit* program [Wheeler, 1990]. In Wheeler’s formulation, information is primary; in RefG, the physical foundation-medium is primary, while the informational trace is a result of its self-distinction.

Here, *boundary* denotes the domain within which a measuring procedure is physically defined. An observer internal to the Universe can measure only those relations that are defined within the active nodal network itself.

Kurt Gödel's incompleteness theorems [Gödel, 1931] provide a fundamental epistemological analogy: just as a consistent formal system contains true propositions that cannot be proved from within that system, a self-measuring Universe raises the question: *How completely can an internal component of the system measure the global boundaries of the entire system?*

This self-measuring picture gives rise to three operational boundaries:

1. **The Operational Origin of Time:** In the selected scenario, time begins with the first foundation-wide self-distinction, and this boundary is assigned  $t = 0$  and  $\tau = 0$ . No physical past exists before the birth of the Universe, because change and the ordering of events are not yet defined. Background parameters determine the post-Genesis history, while the unified bridge among photons, atomic standards, and operational geometry makes that history accessible to an internal observer. The complete process age is finite; its numerical value is fixed by the background parameters, Genesis matching, and clock calibration.
2. **The Operational Boundary of Space:** We measure the connections among distinguishable oscillon nodes. An exterior of the Universe is not a possible location of observation. The cosmological horizon is a measurement boundary defined from within the network. The quantity  $R_{\text{act}}$  is a local activation or correlation scale and does not denote an external radius of the Universe.
3. **The Operational Boundary of Scale:** In RefG, the Planck length,  $\ell_{\text{Pl}} \sim 10^{-35}$  m, is the natural transition scale constructed from  $c$ ,  $\hbar$ , and  $G$ , at which quantum and gravitational effects become simultaneously important. The exact relation between the minimum operational distinguishability threshold  $\ell_*$  and the Planck length must be determined by the microdynamics of the foundation.

In all three cases, the boundary arises from the conditions of measurement internal to the Universe. A finite object may extend to infinity in a chosen coordinate representation, as in the stereographic projection of a sphere onto an infinite plane or in Benoit Mandelbrot's coastline paradox [Mandelbrot, 1982].

### 6.3 Methodological Principles

The methodological foundation of this research program is absolute: **An intuitive physical picture must be translated into a rigorous, computable, mathematically self-consistent, and falsifiable formalism.**

The theoretical analysis of a particle oscillon follows a rigorous 9-step verification chain:

1. **Empirical Grounding:** Precise determination of mass, charge, spin, lifetime, and scattering form factors;
2. **Mode Identification:** Selecting the appropriate resonant and topological class within the foundation-medium;
3. **Geometric Ansatz:** Formulating the spatial geometry, topological invariants  $(Lk, Wr, Tw)$ , and boundary conditions;
4. **Dynamical Law:** Deriving the nonlinear field action, Lagrangian density, and equations of motion;
5. **Symmetries and Conservation Laws:** Identifying Noether currents, charge, spin, color, and zero modes;
6. **Existence and Stability:** Proving the existence of localized finite-energy soliton solutions and analyzing their perturbation spectrum;



7. **Parameter Closure:** Fixing all free coefficients independently of the data to be predicted;
8. **Novel Predictions:** Computing independent, measurable physical observables and their uncertainties;
9. **Empirical Confrontation:** Testing predictions against observational data under pre-defined falsification criteria.

A working hypothesis and a verified computational result are strictly distinguished. Every calculation is tied to reproducible code, explicit datasets, and executable verification gates.

## 6.4 Where This Program Could Die (The Falsification Map)

The hallmark of a legitimate scientific theory is its vulnerability to empirical refutation (Popper’s criterion). RefG establishes a transparent, three-tiered falsification map:

### 1. The Entire Framework is Falsified if:

1. The oscillon sector fails to recover effective Lorentz invariance and spatial isotropy of light speed ( $|\Delta c/c| > 10^{-17}$ ; Herrmann et al. 2009), or gravitational-wave propagation speed deviates from  $c$  ( $|v_{\text{gw}} - c|/c > 10^{-15}$ ; Abbott et al. 2017).
2. At the same fixed total energy, inertial and gravitational mass exhibit a composition-dependent discrepancy exceeding observational bounds ( $\eta > 10^{-15}$ ; MICROSCOPE: Touboul et al. 2022).
3. The global pressure evolution of the foundation-medium fails to yield a unique relation between process and metric time, or the cosmology built from that relation fails to recover the observed CMB, BBN, photon redshift, and Type Ia supernova  $(1+z)$  time-dilation relations [Blondin et al., 2008].
4. Any dimensionless physical ratio changes under a local pressure shift—for example, if a reliably established nonzero  $\Delta\alpha/\alpha$  is found ( $|\Delta\alpha/\alpha| > 10^{-6}$ ; Kotuš et al. 2017).
5. Self-distinction of the foundation fails to establish a unified bridge to mutually distinguishable modes, local causality, and additive accounting of energy, momentum, and charge.

### 2. An Individual Sector is Falsified if:

1. Gravitational-wave energy flux fails to reproduce binary pulsar orbital decay to a precision of  $10^{-3}$ – $10^{-4}$  [Kramer et al., 2021], or scalar breathing modes exceed LIGO-Virgo-KAGRA bounds.
2. The galactic vortex framework fails to reproduce SPARC/RAR data [McGaugh et al., 2016] or the  $8\sigma$  separation between lensing centroids and X-ray gas in the Bullet Cluster [Clowe et al., 2006].
3. Combined vortical-nodal sources fail to recover primary CMB anisotropies, the BAO  $100 h^{-1}$  Mpc ruler, and large-scale structure statistics [Eisenstein et al., 2005].
4. The compact regular-core model fails to match Event Horizon Telescope shadow diameters, the compact-branch oscillation spectrum, or gravitational-wave echo limits [Cardoso and Pani, 2019].
5. The neutral phase-density realization fails to admit an independent cyclic coordinate  $\theta_C$ , the common phase shift cannot be maintained as an exact symmetry, or the mean of  $n_C$  cannot be identified with the cosmological  $\eta_F$ .

### 3. A Specific Geometric Reading is Falsified if:

1. The electron oscillon profile fails to maintain a pointlike electromagnetic form factor down to  $10^{-18}$  m [Navas et al., 2024].
2. The Koide/ $C_3$  branch fails to construct a radiative bridge to pole masses, or a fourth sequential charged lepton is discovered.
3. Galactic-flow circulation is not quantized in units of  $h/m$  within an independently fixed effective-mass and coherence window.
4. The Möbius double-cover fails to yield the Dirac value  $g = 2$ , Fock-space antisymmetry, or the corresponding quantum corrections.
5. The calculated fluctuation spectrum of one oscillon fails to exhibit common  $\Omega_t$  factorization; in that case, universal particle transposition and the mass ratios inferred from it are removed from the program.

Each falsification criterion is tied to specific experimental measurements: an adverse result eliminates the corresponding hypothesis from the research program rather than prompting ad-hoc verbal revisions.

## Conclusion

This monograph has established the unified conceptual architecture of **Refractive Gravity** (RefG):

1. **Ontological Starting Point:** In the Genesis scenario of RefG, one universal pre-spatial foundation undergoes a centerless initial transition that produces the first distinguishable relations. Resonant fields of oscillon seeds localized within this initial relational layer overlap; the resulting coupling increases network connectivity, while stable matter forms through later locking.
2. **Matter as Oscillons:** Locally confined, topologically stable nonlinear wave configurations of the foundation—**oscillons**—induce a static pressure deficit in the surrounding substrate via Bernoulli-type dynamics ( $-\Delta P \propto \rho_{\text{dyn}}$ ), registered by external observers as gravitational mass.
3. **Gravity as Refraction and Coframe Geometry:** A pressure deficit creates the scalar refractive readout. The leading action of the post-Genesis relational-coframe branch takes TEGR form, and the exact torsion-curvature identity converts it to the Einstein–Hilbert action up to a boundary term. Variation of  $\theta_C$  determines phase-current conservation, while metric variation of the phase-current sector gives the Hilbert stress–energy tensor  $T_{\mu\nu}^C$ . On the retained one-metric 1PN branch with universal minimal matter coupling and no additional unsuppressed mode, this geometry gives the complete standard PPN vector; the static, spherically symmetric calculation supplies a separate local check.

This unified pressure chain unfolds across three primary astrophysical scales:

- **Locally:** In compact relativistic objects, where internal topological content and externally read mass decouple via two-channel accounting;
- **Galactically:** In the coherent vortical circulation and polarization of the foundation, which RefG connects to MOND and Tully–Fisher (RAR/BTFR) regularities without dark matter particles;
- **Cosmologically:** From a global centerless boundary of origin, expansion of the foundation lowers its background pressure; that decline reorganizes the material standard and externally read mass. The organized neutral collective phase charge is distributed over the enlarging three-dimensional measure, producing the homogeneous relaxation law. Internal observers read the entire history through the single geometry  $A = a/p$ : redshift, signal time dilation, and distances arise from

that geometry and, on the ideal homogeneous background, match the spatially flat  $\Lambda$ CDM observables at equal parameters. A stationary manifested vacuum phase with constant energy density contributes  $\Lambda_{\text{phase}}$  once to  $\Lambda_{\text{eff}} = \Lambda_{\text{bare}} + \Lambda_{\text{phase}}$ , with a positive total producing late operational acceleration.

At the same time, the volumetric resonator, common transposition, and  $C_3$  structure form a map of particle discreteness. The eigenspectrum of one finite-energy oscillon provides the physical test: it must reveal whether common-cadence factorization and a dynamical connection to Koide's three-phase geometry arise from the particle dynamics itself. The topological invariants  $Lk = Wr + Tw$  remain a candidate geometric language for spin, charge, and color.

Refractive Gravity presents the Universe as a self-measuring resonant system: the state of the foundation creates geometry, geometry shapes the motion of matter and light, and clocks and rulers made from that same foundation read the entire process from within. This monograph brings gravitation, cosmological evolution, the structure of matter, and the origin of measurability into one physical language.

## Current Status of the Research Program

The current status of each theoretical component is cataloged here to maintain clarity and transparency across the research program:

1. **Ontological Foundation, Cosmic Genesis, and the Origin of Measurability**—The current Genesis scenario combines a centerless initial boundary common to the entire foundation, the first minimally distinguishable relations, localization of oscillon seeds, overlap of resonant fields, coupling-mediated network connection, and subsequent stabilization. Formal checks separate the parameter domains, the causal consistency of the event ordering, and the conditional balance of energy-transfer signs after the initial event. A finite periodic graph demonstrates compatibility between connectivity and the absence of a unique center; in a symmetric network, strict radial ordering from one site creates a preferred center and lies outside the centerless domain. Derivation of the physical Genesis mechanism requires the transition action, actual dimension and topology, an oscillon-seed solution, the resonant-field equation, the physical coupling threshold, the initial energy distribution, the thermal history, and observational consequences.
2. **Inter-Node Separation and Minimum Distinguishability**—The operational concept of distinguishability has been established; a Planck-order threshold is a physical candidate, but it has not been derived from the underlying action.
3. **Local Collective Cadence**—The geometric response  $\Omega_t/\Omega_{t,0} = p$  and  $d\tau = p dt$  provides the exact two-time dictionary. The particle factorization  $\nu_i^{(t)} = s_i \Omega_t$  is an independent candidate whose decisive test is the fluctuation operator of one finite-energy oscillon.
4. **Oscillon Deficit, Scale, Time, and Mass**—Metric filters and the first-order biconformal branch have been derived symbolically; a finite-energy nonlinear oscillon profile and the full mass-readout map remain open.
5. **Relational Geometry, the Einstein–Hilbert Action, and 1PN**—In the static, asymptotically uniform, nonrotating, spherically symmetric weak field, the common clock–ruler response gives  $ds^2 = p^2 c^2 dt^2 - p^{-2} d\sigma^2$ ,  $c_{\text{local}} = c_0$ , and  $\gamma = 1$ . With a canonical gradient term, linear coupling to the active Gauss source, and no additional unsuppressed term at the same order, the additive logarithmic response  $u = -\ln p$  gives  $u = GM/(c^2 r)$  and  $\beta = 1$ . The post-Genesis continuum description assumes a  $3 + 1$  relational coframe, one operational metric, flat inertial transport, a leading local and reversible constant-coefficient parity-even quadratic-torsion action, minimal phase coupling, no unsuppressed mixed operator at that order, and the inertial-gauge character of

orientation. In the regular linear Minkowski limit of this class, the TEGR coefficient ratios follow, and the torsion–curvature identity converts the action to Einstein–Hilbert form up to a boundary term. On the positive, stable, causal constitutive branch,  $\rho_C > 0$ ,  $\rho'_C > 0$ , and  $0 \leq n_C \rho''_C / \rho'_C \leq 1$ ; metric variation gives  $T_{\mu\nu}^C$ , while variation of  $\theta_C$  gives a conserved current whose homogeneous reduction recovers  $\eta_F a^3 = 1$ . The Fierz–Pauli/Deser spin-2 route independently checks the Einstein–Hilbert operator. The complete standard PPN vector follows conditionally through 1PN order when the full action is Einstein–Hilbert plus matter universally and minimally coupled to the same metric, RefG and GR use the same source, PPN gauge, background, and boundary conditions, and no additional unsuppressed field or operator enters at this order:  $\gamma = \beta = 1$  and  $\xi = \alpha_1 = \alpha_2 = \alpha_3 = \zeta_1 = \zeta_2 = \zeta_3 = \zeta_4 = 0$ . Open microscopic origins include the node graph and dimension, the smooth coframe and its Lorentzian temporal direction, flat transport and the common cone, the inertial-gauge character of orientation, the coframe–phase split, the oscillon  $T_{\mu\nu}$ , the microphysics of  $\rho_C(n_C)$  and its relation to  $P_F$ , and the numerical values of  $G$  and  $\Lambda_F$ . Also open are 2PN, strong fields, particle-specific source matching, quantitative control of the discrete-to-continuum limit, and the microscopic origin of suppression of mixed, nonminimal, and higher-derivative operators.

6. **Inertia and Gravitational Mass**—The Noether identity is conditionally established for finite-energy, Lorentz-covariant localized solutions satisfying the Laue boundary condition; an exact derivation of the zero mode remains open.
7. **Gravitational Waves and Local Unobservability**—Under the stated continuum premises, the TEGR-equivalent one-metric geometry gives the two standard transverse-traceless (TT) polarizations on the common light cone and the corresponding detector response. The node-scale origin of the common cone, the two TT modes, and the inertial-gauge character of orientation remains open. Cosmological propagation, source–flux normalization, the complete waveform, and microscopic exclusion of possible additional polarizations also remain open.
8. **Compact Objects**—A static, conditional map and finite-core matching of the  $C^2$  ansatz have been obtained; within one specific exterior-layer and pressure-reversal criterion, admissible domains do not overlap. The physical equation of state, gravitational collapse, rotation, and global stability remain open.
9. **Galaxy Rotation**—An empirical test using  $B - V$  colors did not detect the anticipated age correlation. This result is compatible with common rescaling, but it does not establish either the Doppler-cancellation mechanism or a complete physical explanation of the MOND correlations; both remain open. The symbolic algebraic cancellation  $c_{\text{coord}}/c_0 = p^2$  has been obtained, but direct physical confirmation from PHANGS/SPARC data and an exact process-level derivation of vortex rotation are still lacking.
10. **Cosmological Relaxation, the Phase-Current Source, and the Time Bridge**—The ratio  $A = a/p$  is the internal readout of one expansion–pressure–standard history. In the post-Genesis homogeneous description,  $\Psi_C = \sqrt{n_C} e^{i\theta_C}$  represents a neutral collective phase state, and phase-current conservation on a comoving three-dimensional measure gives  $\eta_F a^3 = 1$  exactly. In the covariant action for a barotropic, isentropic, irrotational phase current, variation of  $\theta_C$  gives  $\partial_\mu J^\mu = 0$ , the barotropic energy function defines  $p_C = n_C \rho'_C - \rho_C$ , and metric variation gives the Hilbert stress–energy tensor  $T_{\mu\nu}^C$ . Homogeneous constitutive relations yield  $P_F/P_{F,0} = a^{-3}$ ,  $p = a^{-3/2}$ , and  $A = a^{5/2}$ . The same  $A$  determines redshift, signal dilation, and distances and is ideally operationally equivalent to spatially flat  $\Lambda$ CDM at equal background parameters. A constant vacuum contribution enters one effective  $\Lambda_{\text{eff}}$  term. Open problems are the node/oscillon origin of  $\rho_C(n_C)$ , the relation of  $p_C$  to  $P_F$ , the microscopic coframe–phase split, particle-specific sources, concrete inhomogeneous solutions, the numerical value of  $\Lambda$ , Genesis matching, and a complete data model. Catalogue preference has not been established.

11. **Two-Level Time and Quantum Randomness**—The interpretive framework is compatible with Bell tests and delayed-choice experiments; formal derivations of Born’s rule, the no-signaling condition, and measurement dynamics remain open.
12. **Particles as Oscillon Modes**—The current targeted stage comprises one oscillon family: its local dynamics, a finite-energy solution, the fluctuation operator, a test of common frequency factorization and of the  $t/\tau$  dictionary, followed by an independent energy-to-rest/pole-mass map.
13. **The Koide/ $C_3/2/9$  Structure**—In  $x_k = \sqrt{m_k}$  space, the  $C_3$  algebra yields  $K = 2/3$  exactly when  $A_K = \sqrt{2}$ ; with  $(A_K, \theta) = (\sqrt{2}, 2/9)$  anchored to the electron pole mass, the tree-level values miss the muon by  $451.7\sigma$  and the tau by  $0.61\sigma$ . The  $C_3$  structure remains an external algebraic candidate; its physical realization must arise from oscillon dynamics, its eigenspectrum, and a radiative pole-mass bridge.
14. **Spin, Pauli Exclusion, and Fermionic Statistics**—The Möbius double-covered geometry provides a candidate mechanism; a complete derivation of Fock space and the Spin-Statistics Theorem from first principles remains open.
15. **Charge, the Photon, and the Electromagnetic Channel**—The following conditional structural results have been obtained: local phase  $U(1)$ , two helicities, screening of the hedgehog’s otherwise unscreened asymptotic sigma-field component, whose energy grows linearly with radius, together with the surviving  $1/r^2$  far field and flux accounting after screening, and subtheorems for a conserved phase current. Canonical charge normalization, Maxwell’s equations, Gauss’s law, Ward identities, quantum electrodynamics, the Dirac sector, the exact Hilbert source of freely propagating radiation, and its pressure-deficit geometric response remain open.
16. **Color, Quarks, and Gluons**—The three-color, eight-mode construction provides only a partially compatible structural map;  $SU(3)$  structure constants, interaction vertices, scale evolution, the confinement potential, and fractional charge normalization remain open.
17. **Electroweak Map and the Higgs**—This constitutes a geometric placement map;  $SU(2)_L$  chirality, a  $W/Z$  mass matrix preserving a massless photon, CKM/PMNS mixing, neutrino mass hierarchy, Yukawa couplings, and the vacuum expectation value remain open.
18. **The Fine-Structure Constant  $\alpha$** —A nodal interpretation and target-independent spectral measure have been obtained:  $\alpha$  is read from the relative strengths of the electric-flux and transverse-photon response channels, with the overall scale canceling. A microscopic action for physical charge  $Q$ , a unique numerical value, the complete charged spectrum, and RG flow to the Thomson limit remain open. The earlier boundary formula remains a target-dependent conditional result.
19. **The Gravitational Hierarchy**—The force ratio is expressed algebraically as  $F_{\text{em}}/F_g = \alpha(M_{\text{Pl}}/m)^2$ ; the fundamental origins of  $G$ ,  $M_{\text{Pl}}$ , the oscillon energy scale, and the mass hierarchy remain open.
20. **A Self-Measuring Manifested Universe**—The global-resonator picture is a cosmological-level hypothesis. A finite periodic graph accommodates connectivity and the absence of a unique graph-theoretic center within one mathematical construction. Cosmic finiteness, boundarylessness, physical dimension, and topology lie beyond this example; low-multipole and topological signatures remain candidate observables.

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